Machine designation:	Structural-steel rolling mill
Machine type	Rail mill
Machine number:	0100133
Order No.:	94-630000
Year of construction:	2012
Customer:	SAIL, Bhilai Steel Plant

Rev.	Issue date	Type of revision
00	06.02.2012	Original version
01	22.02.2012	Editorial revision
02		
03		
04		
05		

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4 Safety layout

The machine must not be operated unless all danger-zone safeguarding elements are in place and intact.

The required danger-zone safeguarding elements (railings, barriers, covers, etc.) and all warning and instruction signs are shown in the safety layout.

The foundation covers, grids, fences, barriers and fire extinguishing equipment, etc., shown in the safety layout are not included in the SMS Meer GmbH standard scope of supply of the machine. They are, however, absolutely necessary for safe operation and must be installed by the user or the persons entrusted by him with this job, before commissioning.



Since there is an extremely high risk of getting injured in the immediate surroundings of the rail rolling mill, but on the other hand persons need to be present, this risk must be minimised by accessory measures.

Operator instructions on the subjects of safety and behaviour at the groove dressing machine, which are given at regular intervals (once every year), as well as regular inspections/ observations of production are mandatory.

All actions for minimising the potential dangers that have been described and shown here must be carried out prior to commencing production.

4.1 Machine / plant data

Machine designation:	Structural-steel rolling mill
Machine type	Rail mill
Machine number:	0100133
Order No.:	94-630000
Year of construction:	2012
Customer:	SAIL, Bhilai Steel Plant

4.1.1 Persons/companies involved

The safety layout has been prepared by:

Company: SMS Meer GmbH

Name / Dept.: Prätzel, Alfred
(Documentation + Product Safety)

Department: 675300

Signature: .....

The specialist departments, persons and signatories indicated below were involved in the preparation of the safety consideration.

Rev.	Name:	Company:	Department:	Dept. No.:	Date:	Signature:
01	Mr. Reismann	SMS Meer	Engineering Section and Billet Mills	686300	26.03.12	
	Mr. Jansen		Engineering Section and Billet Mills	686300	26.03.12	
	Mr. Gonsior		Electrical Systems Long Products	686700	26.03.12	
	Mr. Jaek		Hydraulic Systems Long Products	686300	26.03.12	

4.2 Definition of plant components/sub-sections:

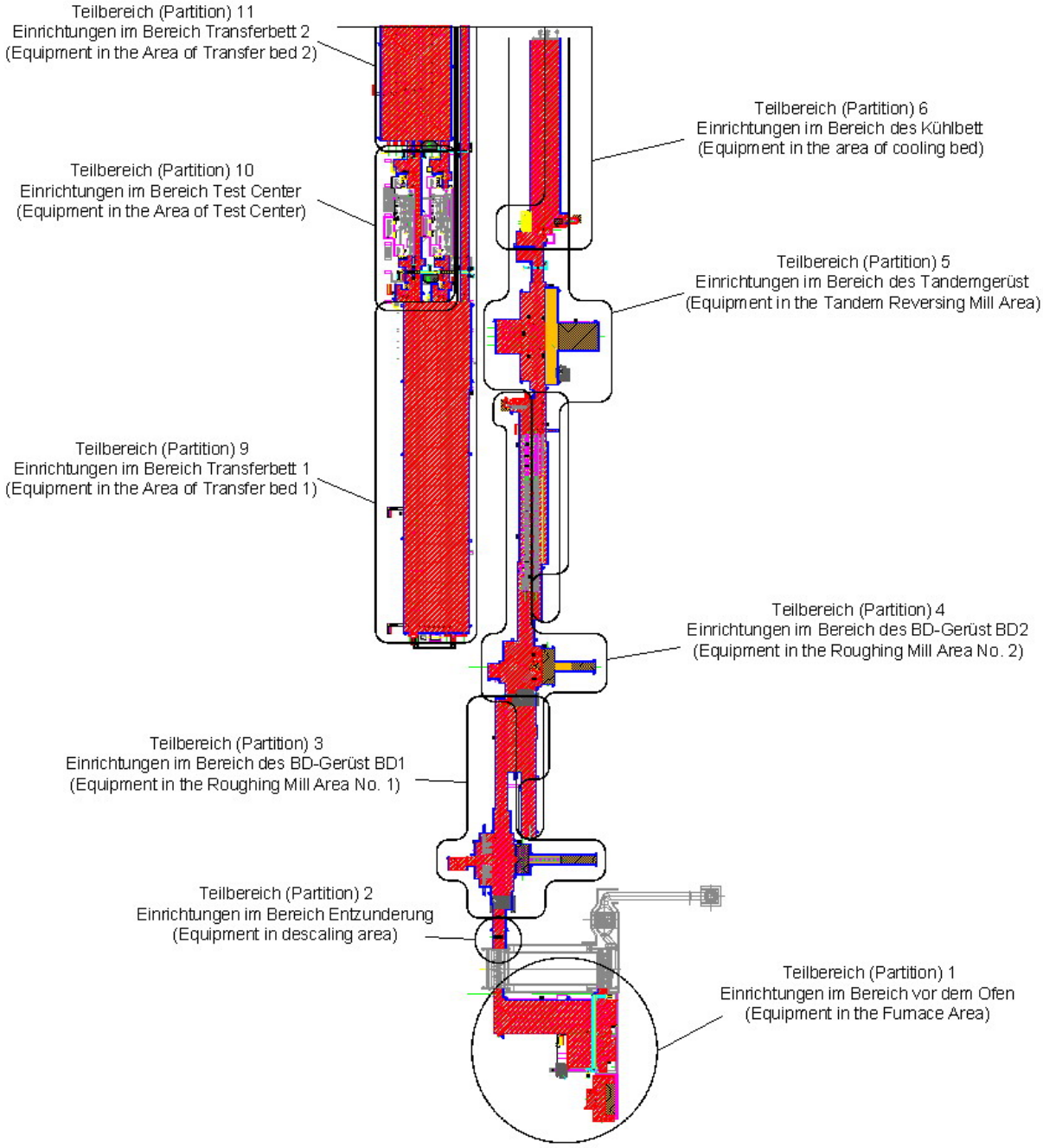


Figure 4-1 Definition of plant components/sub-sections (Part 1)

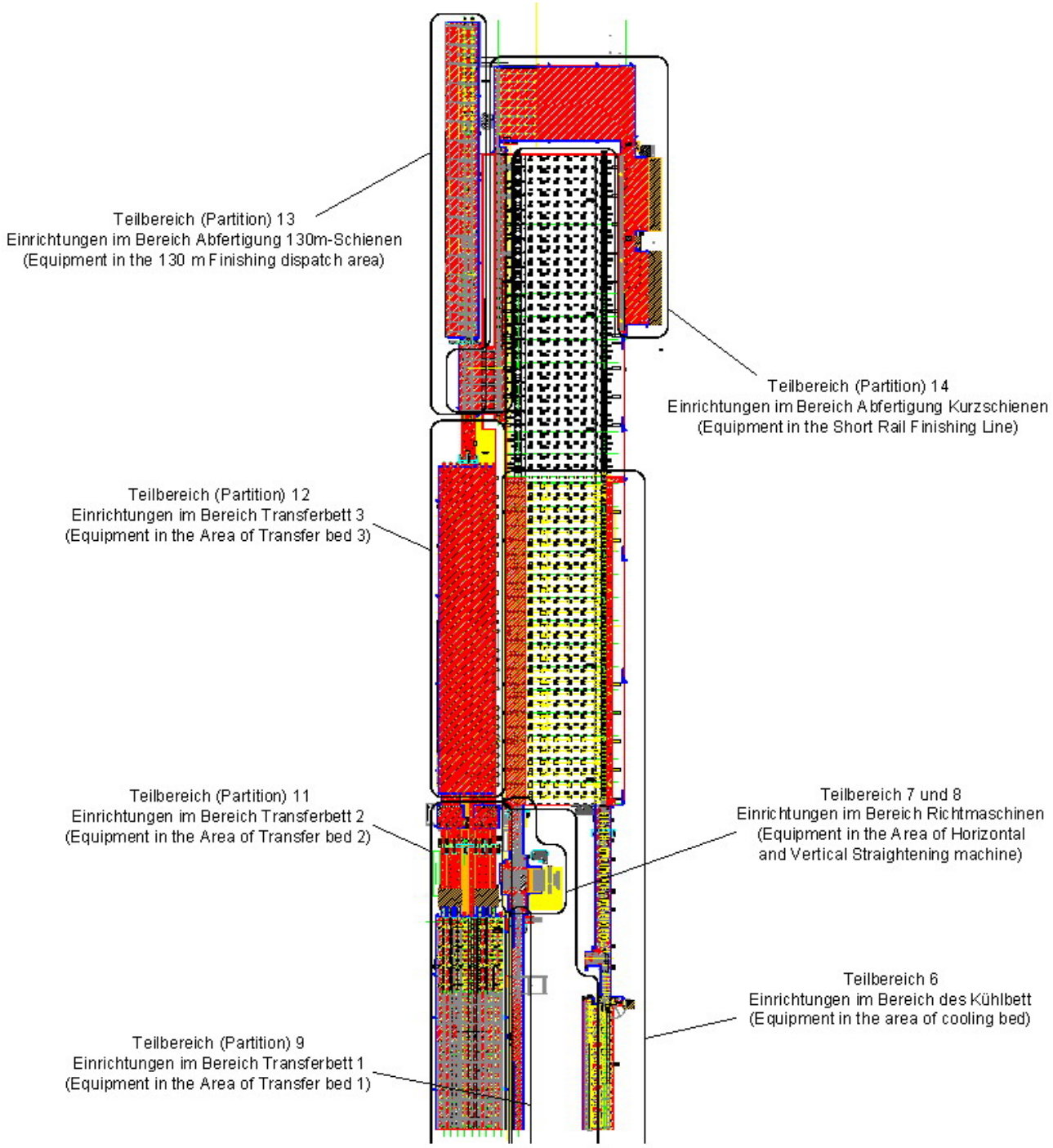


Figure 4-2 Definition of plant components/sub-sections (Part 2)

4.3 Danger zones (definition of sub-sections)

4.3.1 Danger zones/sub-sections of the plant

		Drawing number
Sub-section 1	**Equipment in area upstream of furnace**	
	(Layout)	0100133-00.650_01
	(Layout of slab transfer car area)	0100133-00.650_02
Sub-section 2	**Equipment in area of descaler**	
	(Layout)	0100133-00.651_01
Sub-section 3	**Equipment in area of stand BD1**	
	(Layout)	0100133-00.651_01
	(Horizontal section)	0100133-00.651_02
Sub-section 4	**Equipment in area of stand BD2**	
	(Layout)	0100133-00.652_01
	(Horizontal section)	0100133-00.652_02
Sub-section 5	**Equipment in area of tandem mill stand**	
	(Layout)	0100133-00.653_01
Sub-section 6	**Equipment in area of cooling bed**	
	(Layout)	0100133-00.654_01 0100133-00.654_02 0100133-00.654_03
	(Horizontal section)	0100133-00.654_04 0100133-00.654_05
	(Sections)	0100133-00.654_06
Sub-section 7 + 8	**Equipment in area of straighteners**	
	(Layout)	0100133-00.655_01
	(Horizontal section)	0100133-00.655_02
	(Detail)	0100133-00.655_03
Sub-section 9	**Equipment in area of transfer bed 1 upstream of test centre**	
	(Layout)	0100133-00.656_01
Sub-section 10	**Equipment in area of test centre (and cold saws 1 to 4)**	
	(Layout)	0100133-00.657_01
Sub-section 11	**Equipment in area of transfer bed 2 (and visual inspection area and cold saws 5 to 10)**	
	(Layout)	0100133-00.658_01
Sub-section 12	**Equipment in area of transfer bed 3 (and straightening presses)**	
	(Layout)	0100133-00.659_01
	(Sections)	0100133-00.654_06
Sub-section 13	**Equipment in the processing area for 130 m rails**	
	(Layout)	0100133-00.660_01
	(Horizontal section)	0100133-00.660_02

Sub-section 14	**Equipment in the processing area for short rails**	
	(Layout)	0100133-00.661_01
	(Horizontal section)	0100133-00.661_02

4.3.2 Danger zone - hydraulic, pump and utility rooms, scale flumes

The doors to the hydraulic, pump, utility and electrical switchgear rooms and to the scale flumes are of **lockable** (**locked**) design.
The doors must always remain **closed** during production.

Access to these areas is allowed only to **trained personnel** who have been **authorised** by the plant owner**.**

The keys are kept in the respective control room of the allocated plant section.

The owner must assign the keys to the designated persons, and the authorisation to hold keys must be stipulated in the operating and work instructions.

The access doors must be locked again after completion of the work and when all persons have left the danger zone.
If necessary, the keys must be again deposited in the control room.

The plant may be started only after the plant operator has made sure that all doors have been locked again and all keys deposited in the control room.



Even with the power supply switched off and hydraulic systems depressurised, access to the danger zones is permitted only to specially trained persons.

Access to the danger zone must be limited to those persons who are directly engaged in the operation of the beam rolling mill and its auxiliary equipment.

The authorised persons shall be appointed by the owner.



All doors must be equipped with an escape unlocking mechanism on the inside.

The pump stations and valve stands must be equipped with catch trays for the collection of leak oil.

4.4 Signs on the machine

The places at which the signs are installed are marked by item numbers (number in circle).

Warnings and instruction signs are arranged at the machine/plant at easily visible places.

Damaged, illegible and lost signs must be replaced promptly.

Signs on components relevant to safety (pressure accumulators, load-bearing structural components) must be affixed using adhesive only.

Sign 1

Material No. 1025102 MMN 182.02

Quantity: n



Sign 2

Material No. E106 456 MMN 182.03

Quantity: n



On all doors to the danger zone.
Installation position not shown in safety layout.

Where the wearing of personal protective equipment is governed by works rules, no signs need to be mounted.

Sign 3

Material No. 1021386 MMN 180.02

Quantity: n



Located at all control cabinets, control panels and terminal boxes

Sign 4

Material No. E289 357 SEN 772 41

Quantity: n



In the area of the chip and scrap container and at the loading beds

Sign 5

Material No. E307 112 SEN 772 35

Quantity: n



In the area of the rolling mills, straightener and roller tables

Sign 6

Material No. E106 204 SEN 772 20

Quantity: 1



Sign 7

Material No. E498 184 SEN 772 92

Quantity: 1



Sign 8

Material No. 1021382 MMN 180.01

Quantity: n



Sign 9

Material No. E285 979 SEN 772 31

Quantity: 4



Sign 10

Material No. 1024806 MMN 181.04

Quantity: n



On the pressure accumulator,
on the hydraulic tank
installation position not shown in the safety layout

Sign 11

Material No. 1026185 MMN 181.07

Quantity: n

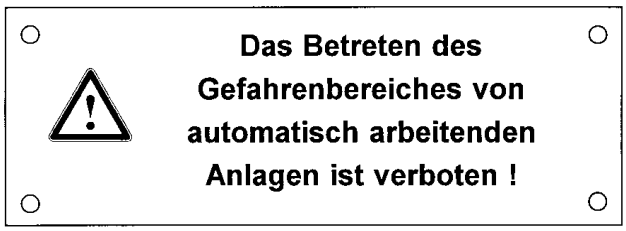


On all doors leading to the danger zone
and machine cellar,
at the stairs to the work platform,
hydraulic room, electrical switchgear room

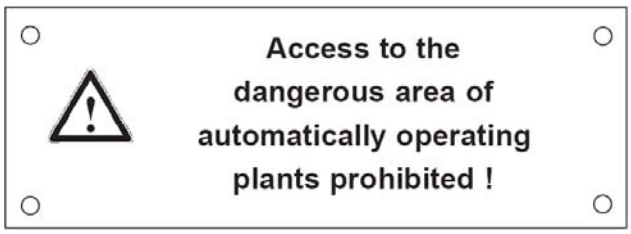
Sign 12

SEN 772 27 Quantity: n

Material No. E296 592 (DE)



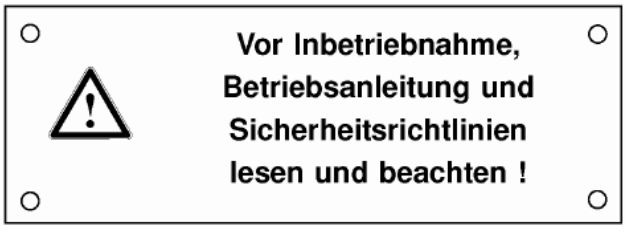
Material no. E318 998 (EN)



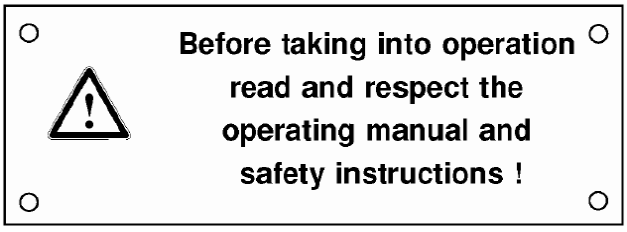
Sign 13

MMN 182.04 Quantity: n

Material No. 106,180 (DE)



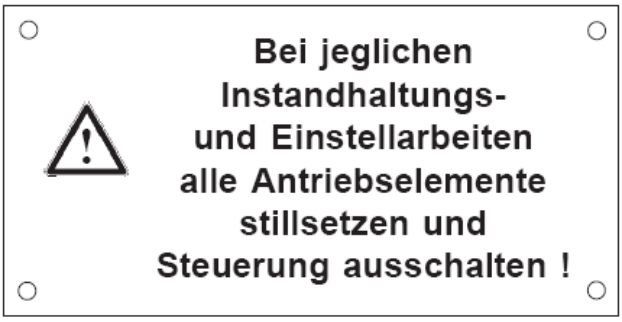
Material No. 106,181 (EN)



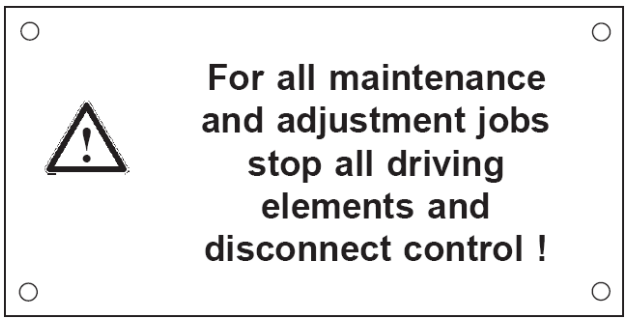
At all operator stations

Sign 14 SEN 772 22 Quantity: n

Material No. E102 251 (DE)

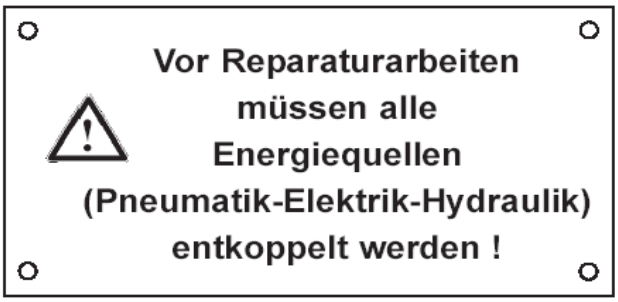


Material No. E102 252 (EN)

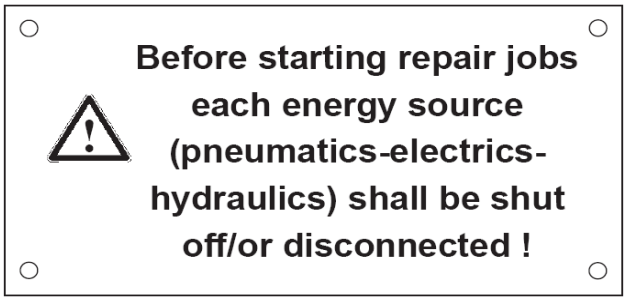


Sign 15 MMN 182.01 Quantity: n

Material No. 1024756 (DE)

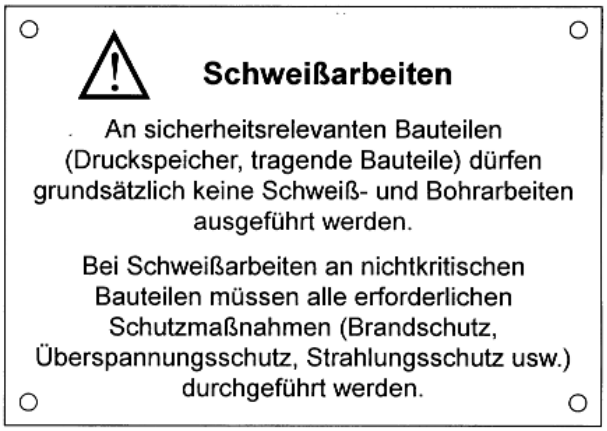


Material No. 1024757 (EN)

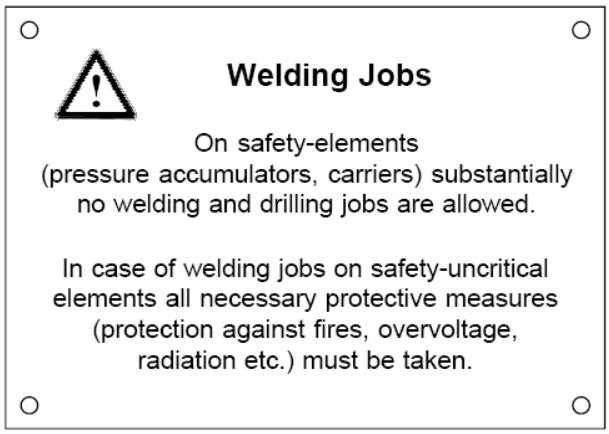


Sign 16 MMN 181.01 Quantity: n

Material No. E283 303 (DE)



Material No. E283 317 (EN)



On the pressure accumulator, on the hydraulic tank/unit

Sign 17 SEN 772 25 Quantity: n

Material No. E215 957 (DE)



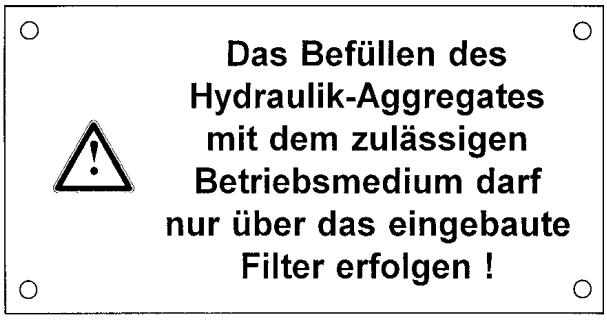
Material No. E240 120 (EN)



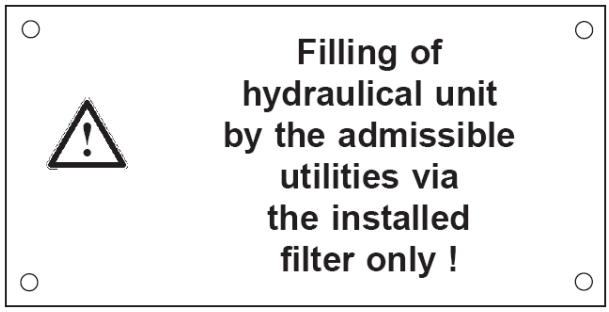
Directly on the pressure accumulator

Sign 18 MMN 182.05 Quantity: n

Material No. 1024897 (DE)



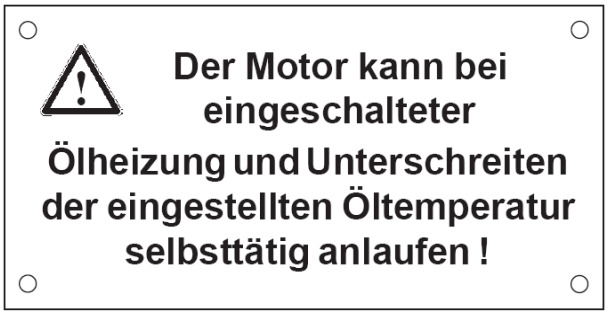
Material No. 1024899 (EN)



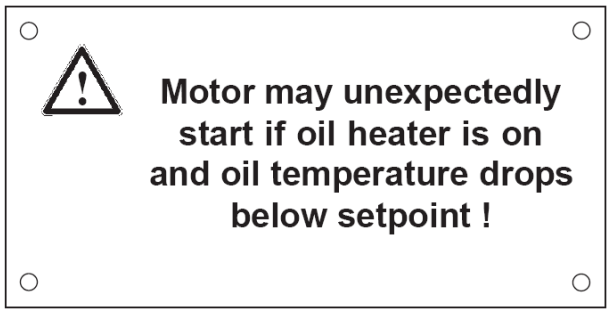
Directly on the hydraulic unit, next to the filling opening (connection).

Sign 19 MMN 180.03 Quantity: n

Material No. 1022098 (DE)

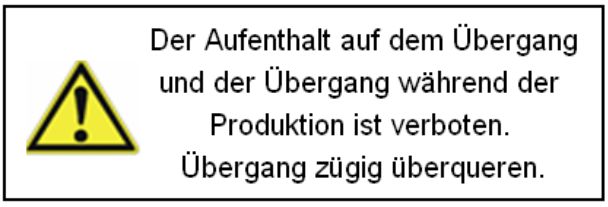


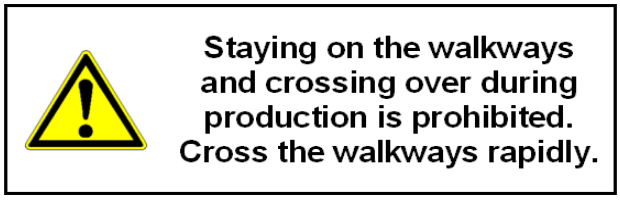
Material No. 1022100 (EN)



On the hydraulic unit

Sign 20 Quantity: n





At all walkways

Sign 21 MMN 376.38 Quantity: n

Material No. 1024122 (DE)



Material No. 1024127 (EN)

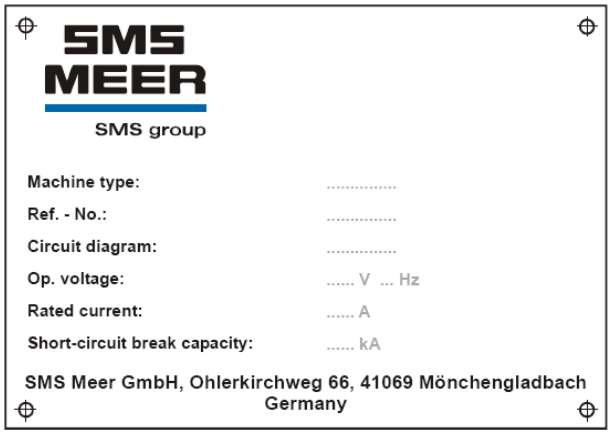


Sign 22 MMN 183.02 Quantity: n

Material No. 1024049 (DE)



Material No. 1024052 (EN)



On all control cabinets

Sign 23

Material No. 1024810 MMN 183.02

Quantity: n



In area of induction heating system

Sign 24

Warning sign: D-P016 DIN 4844-2

Quantity: n



In area of induction heating system

Sign 25

Warning sign: D-W012 DIN 4844-2

Quantity: n



In area of induction heating system

4.5 Definition and relationships of operating modes, functions and operating situations

Operating mode	Functions	Operating situation
Automatic mode	In the Automatic mode, the individual program steps are combined into a fully automatic cycle. The intervention of persons in the danger zone is not required and is prohibited. The course of sequences (successive and/or parallel) is performed entirely via a control unit (e.g. PLC). Control sequences start without operator interventions. The operator monitors the process only.	
	Production **Changing of the rolls, mill stand, straightening rollers and tools/operating materials**	**No persons** in the danger zones.
		Safety equipment active.
		All **doors** to the aforementioned danger zones are closed and **locked** and/or **electrically locked (with guard locking)**.
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**). **1)**
	Calibration with active safety system	**No persons** in the danger zones.
		Safety equipment active.
		All **doors** to the aforementioned danger zones are closed and **locked** and/or **electrically locked (with guard locking)**.
		There is no material in the production line.
		The furnace outlet is blocked, i.e. no new material comes into the plant.
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**). **1)**
Semi-automatic mode	This operating mode is identical with the Automatic mode, except that only individual cycles are performed, i.e. after an individual cycle has been performed, the next program step has to be started again manually. All mechanical and technological interlocks as well as the speed change are effective. The individual partial functions of the automatic sequence can run alone (started via pushbutton). No operation may be performed automatically on switching to the 'Manual' mode. Movements must be re-initiated. Mechanical and technological interlocks must be taken into account.	
	Production **Changing of the rolls, mill stand, straightening rollers and tools/operating materials**	**No persons** in the danger zones.
		Safety equipment active.
		All **doors** to the aforementioned danger zones are closed and **locked** and/or **electrically locked (with guard locking)**.
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**).
	Calibration with active safety system	**No persons** in the danger zones.
		Safety equipment active.
		All **doors** to the aforementioned danger zones are closed and **locked** and/or **electrically locked (with guard locking)**.
		There is no material in the production line
		The furnace outlet is blocked, i.e. no new material comes into the plant
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**).

Operating mode	Functions	Operating situation
	Calibration with limited safety  The function "Calibration with limited safety" may be selected only by **trained personnel** who have been **authorised** by the plant owner (**with authorisation** via key-operated switch). Improper operation may result in serious injury and damage to the equipment.	
		All machine movements can be performed step by step and one after the other.
		Movement of the machine must be executed under the **supervision** of the operator.
		There is visual contact with the machine or plant component to be controlled as well as (possible) voice contact with persons at the machine/plant.
		Authorised and appointed persons are likely to be present **in** the relevant **danger zone**.
		Part of the safety equipment is not active.
		Some of the **doors** are **not locked**.
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**).
	Inspection (inspection under load)  The "Inspection" function may be selected only by **trained personnel** who have been **authorised** by the plant owner (**with authorisation** via key-operated switch). Improper operation may result in serious injury and damage to the equipment.	
		The "Inspection" function is used for production in the **Semi-automatic** operating mode, at reduced speed (**<3m/s**) and limited safety.
		All machine movements can be performed step by step and one after the other.
		In the **Inspection under load** mode, all machine movements can be executed successively (sequence by sequence) in the jogging mode. The material speeds are reduced (**speeds < 3m/s**).
		Movement of the machine must be executed under the **supervision** of the operator.
		There is visual contact with the machine or plant component to be controlled as well as (possible) voice contact with persons at the machine/plant.
		Authorised and appointed persons are likely to be present **in** the relevant **danger zone**.
		Part of the safety equipment is not active.
		Some of the **doors** are **not locked**.
		Control (start command) is from the main control desk

Operating mode	Functions	Operating situation
Manual mode	In the Manual mode, all machine movements are started and stopped manually. The monitoring devices and interlocks intended for protecting the machines are active. Manual mode allows the execution of all machine functions under increased responsibility of the operator. Faulty operation (faulty operation sequence) cannot be ruled out.	
	Production **Changing of the rolls, mill stand, straightening rollers and tools/operating materials**	**No persons** in the danger zones.
		Safety equipment active.
		All **doors** to the aforementioned danger zones are closed and **locked** and/or **electrically locked (with guard locking)**.
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**).
	Calibration with active safety system	**No persons** in the danger zones.
		Safety equipment active.
		All **doors** to the aforementioned danger zones are closed and **locked** and/or **electrically locked (with guard locking)**.
		There is no material in the production line.
		The furnace outlet is blocked, i.e. no new material comes into the plant
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**).
	Calibration with limited safety  The function "Calibration with limited safety" may be selected only by **trained personnel** who have been **authorised** by the plant owner (**with authorisation** via key-operated switch). Improper operation may result in serious injury and damage to the equipment.	
		All machine movements can be performed step by step and one after the other.
		Movement of the machine must be executed under the **supervision** of the operator.
		There is visual contact with the machine or plant component to be controlled as well as (possible) voice contact with persons at the machine/plant.
		Authorised and appointed persons are likely to be present **in** the relevant **danger zone**.
		Part of the safety equipment is not active.
		Some of the **doors** are not **locked**.
		Control (Start command) is effected from the main control station or from a local control desk (**Local Desk**).

<table>
<tr><th>Operating mode</th><th>Functions</th><th>Operating situation</th></tr>
<tr><td rowspan="11"></td><td colspan="2">Service
 
The "Service" function may be selected only by trained personnel who have been authorised by the plant owner (with authorisation via key-operated switch).
Since the persons authorised to operate the Service function may be others than those authorised to operate the other functions, a separate key-operated switch has been provided for the function.
Improper operation may result in serious injury and damage to the equipment.</td></tr>
<tr><td rowspan="9"></td><td>In the Service function, all machine movements are executed exclusively in the jogging mode. The functions of the safety system are limited.</td></tr>
<tr><td>The pertaining drives remain switched on for as long as the Start pushbutton is pressed (jogging mode).</td></tr>
<tr><td>All machine movements can be performed step by step and one after the other.</td></tr>
<tr><td>Movement of the machine must be executed under the supervision of the operator.</td></tr>
<tr><td>There is visual contact with the machine or plant component to be controlled as well as (possible) voice contact with persons at the machine/plant.</td></tr>
<tr><td>Authorised and appointed persons are likely to be present in the relevant danger zone.</td></tr>
<tr><td>Part of the safety equipment is not active.</td></tr>
<tr><td>Some of the doors are not locked.</td></tr>
<tr><td>Control (Start command) is effected from the main control station or from a local control desk (Local Desk).</td></tr>
<tr><td colspan="2"> 
Movement of the various plant components under the 'Service' function is only possible using the 'enabling pushbutton' to be supplied. During this process, all movements required are performed at reduced speed.
The position of the enabling pushbuttons (e.g. at the access doors or at a "Local Desk") is described in the sections for the respective plant areas.</td></tr>
</table>



1) **Observe:**
Upon activation of the Start button (e.g., in the Automatic mode), the rolling mill might not start up. The rolling mill starts automatically only when the first material is transported from the furnace, provided the safety equipment has been activated.

4.5.1 Relationships between operating modes and functions

Mode: / Function:	Automatic mode	Semi-automatic mode	Manual mode
Production	1	1	1
Changing of the rolls, millstand, levelling rolls and tools	1	1	1
Calibration with active safety system	1	1	1
Calibration with limited safety	0	1 *)	1 *)
Inspection (inspection under load)	0	1 *)	0
Service	0	0	1 *)

1 = applicable, 0 = not applicable

*) Operation at reduced speed (where necessary)

4.6 Access to the danger zones

Warnings and instruction signs referring to the function and hazard potential are provided at all access facilities.



All doors must at all times be able to be opened from the inside. "Escape route".



Access to the danger zone must be limited to persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment and who are able to assess the potential hazards. The personnel must have been trained and instructed in safety issues.

The authorised persons shall be appointed by the owner.

Persons are not allowed to stay in the danger zone.

In the Service mode, opening of the doors is only permitted to specially-trained persons who are familiar with the plant.



When persons are present in the danger zone, it must be ensured via the machine control system that no material from the neighbouring sections can be transported into this danger zone.

Exception: "Inspection under load"

4.6.1 Electrically locked doors

Electrically locked doors are equipped with an electrical guard locking, i.e. the doors shall **only** be opened when the plant has been switched off and in special operating situations.

In case of a power failure all doors can be opened from the inside and outside. This also applies if the Emergency Stop function is activated.

During production the doors must be closed and electrically locked. It is not possible to open the doors.

Exception: Inspection/Inspection under load

4.6.1.1 Opening of an electrically locked door

To open a door, the request button which is arranged in the immediate vicinity of the door in question must be operated. The in-charge operator in the control room receives a signal that the danger zone will be entered. The **plant** is **shut down** by the operator.

The door shall remain locked and cannot be opened again until all hazardous movements have stopped and the operator in the control room "**CP…**" has released the relevant doors.

After the release for door opening has been granted, a lamp near the request button signals that the door is not locked and that the area in question may be entered.



The light indicates:

Continuous light:	Door is **open** The area concerned can/may be entered.
Flashing:	Door is closed but not locked.
No light:	Door is **closed** and **locked**. The area concerned can/may **not** be entered.

Once the door release has been granted, the following interlocks are activated:

- The plant can**not** be started up, all movements are blocked.

  Exception: 'Service' and 'Inspection under load'

- Hazardous movements which are not absolutely necessary cannot be performed.
- Safe speed reductions are activated
  (for: Inspection/Inspection under load).

4.6.1.2 Closing of an electrically locked door (Restart after an operational interruption)

After all persons have left the relevant danger zone and all doors to the relevant danger zone are closed, the person responsible for the action may press one of the acknowledgement buttons located outside next to the relevant door.

Pressing the acknowledgement button signals to the machine operator that no persons are in the danger zone.

The doors to the relevant danger zone are then locked by the machine control system.

The plant may be restarted once the access door has been closed.

It is no longer possible to open the doors.



After the plant has been stopped via the **emergency stop** function and/or **before** renewed **start of production** the plant operator must make sure that **no persons are trapped in the danger zones**.

4.6.2 Light curtain

4.6.2.1 Deactivating a light curtain

To deactivate a light curtain, the request button which is arranged in the vicinity of the light curtain in question must be actuated. The in-charge operator in the control room receives a signal that the danger zone can be entered. The **plant** is **shut down** by the operator.

The light curtain remains active until all hazardous movements have been brought to a standstill.

Once release has been granted, a light in the vicinity of the request button signals that the light curtain is **deactivated** and the area in question may be entered.



The light indicates:

Continuous light:	Light curtain is **deactivated** The area concerned can/may be entered.
No light:	The light curtain is **activated**. The area concerned can/may **not** be entered.

Once the light curtain release has been granted, the following interlocks are activated:

- The plant can**not** be started up, all movements are blocked.

  Exception: 'Service' and 'Inspection under load'
- Hazardous movements which are not absolutely necessary cannot be carried out.
- Safe speed reductions are activated
  (for: Inspection/Inspection under load).

4.6.2.2 Activation of a light curtain (re-start after an operational interruption)

Once all persons have left the danger zone concerned, the responsible person may actuate one of the acknowledgement buttons placed outside the danger zone in the vicinity of the light curtain.

Pressing the acknowledgement button signals to the machine operator that no persons are in the danger zone.

Once the light curtain has been activated, the plant may be restarted.



After the plant has been stopped via the **emergency stop** function and/or **before** renewed **start of production** the plant operator must make sure that **no persons are trapped in the danger zones**.

4.6.3 Access conditions



When persons are present in a danger zone, it must be ensured via the machine control system that no material from the neighbouring sections can be transported into this danger zone.

Exception: "Inspection under load"



Please note:

- When performing maintenance and service work, all drives must be switched off and secured against restarting. Work may be performed only when the master switch has been switched off and secured against restarting.
- When carrying out work on the hydraulic/pneumatic equipment, the power supply (hydraulic/pneumatic power) must be cut off, secured against restarting and, if necessary, the residual energy systematically released prior to performing any work.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.

4.6.3.1 Sub-section 1 (equipment in area upstream of furnace)

4.6.3.1.1 (Deposit area 1) Deposit system for cold blooms

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-B01-01 and D-B01-02</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zone
Deposit system for cold blooms and return system for hot blooms</td></tr>
<tr><td>Access required for:</td><td colspan="6">-</td></tr>
<tr><td rowspan="2">during:</td><td>Servicing/ Maintenance</td><td colspan="2">Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td colspan="2">X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>BC10</td><td colspan="6">- Deposit system for cold blooms</td></tr>
<tr><td>BR10</td><td colspan="6">- Feed roller table for furnace (in deposit area 1)</td></tr>
<tr><td>BP11</td><td colspan="6">- Stopper must be raised</td></tr>
<tr><td></td><td colspan="6">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td colspan="2" rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td colspan="2">-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP1".
Before entering the danger zone, the relevant plant area must be switched off from the CP1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-B01-01 to D-B01-03) must be locked again and
- the keys deposited in the CP1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.1.2 (Deposit area 2) Deposit system for cold blooms and return system for hot blooms

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-B02-01 to D-B02-06</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Deposit system for cold blooms and return system for hot blooms</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Servicing/
Maintenance</td><td>Troubleshooting/
Fault elimination</td><td>Use/Operation
(workplace)</td><td colspan="2">Setting-up
(workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>BR20</td><td colspan="4">- Return roller table for hot blooms</td></tr>
<tr><td>BC30</td><td colspan="4">- Return system for hot blooms
- Deposit system for cold blooms</td></tr>
<tr><td>BZ10</td><td colspan="4">- Weighing system</td></tr>
<tr><td>BR10</td><td colspan="4">- Feed roller table for furnace</td></tr>
<tr><td>BP11 and BP12</td><td colspan="4">- Disappearing stopper</td></tr>
<tr><td>BP31</td><td colspan="4">- Stopper must be raised</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically
locked</td><td colspan="2">Electrically</td><td rowspan="2">with
request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with
request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP1".
Before entering the danger zone, the relevant plant area must be switched off from the CP1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-B02-01 to D-B02-02) must be locked again and
- the keys deposited in the CP1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.1.3 Surface machining area with surface grinder and bloom turning unit

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-B05-01</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zone
Surface machining
with:
surface grinder and bloom turning unit</td></tr>
<tr><td>Access required for:</td><td colspan="6">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td rowspan="3">Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>BA10</td><td colspan="5">- Surface grinder
- Bloom turning unit</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td colspan="2" rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td>X</td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP1".
Before entering the danger zone, the relevant plant area must be switched off from the CP1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-B05-01) must be locked again and
- the keys deposited in the CP1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.2 Sub-section 2 (equipment in area of descaler)

4.6.3.2.1 Descaling system

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D01-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Descaling system</td></tr>
<tr><td>Access required for:</td><td colspan="5">Spray ring change</td></tr>
<tr><td>during:</td><td>Servicing/ Maintenance</td><td colspan="2">Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td colspan="2">X</td><td></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td colspan="5">- Runout roller table for furnace</td></tr>
<tr><td>DR10</td><td colspan="5">- Roller table in area of descaler</td></tr>
<tr><td>DR20, DR30, DR40</td><td colspan="5">- Work and extension roller table upstream and downstream of stand BD1</td></tr>
<tr><td>DH50</td><td colspan="5">- Stand BD1 (main drive)</td></tr>
<tr><td>DW10</td><td colspan="5">- High-pressure water station</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP1".
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Before entering the danger zone:
- the plant area must be switched off from the CP1 control room.
- the high-pressure water station must be depressurised and switched off.
- the shut-off valve on the high-pressure water station must be secure by a lock against re-starting.
- a prohibitive sign "Do not switch on" must be mounted at the shut-off valve of the high-pressure water station.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-D01-01) must be locked again and
- the keys deposited in the CP1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.3 Sub-section 3 (equipment in area of stand BD1)

4.6.3.3.1 Stand BD1 (drive)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D04-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of stand BD1</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
Exception: „Troubleshooting"
DH50 - Stand BD1 (main drive)
- Hydraulic and pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td>-</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP2".
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.

Condition:
- Access to this danger zone is allowed only for authorised and appointed personnel.

- All persons who have entered or left the danger zone must sign in and out in the control room CP2 so that the plant operator is informed about as to whether persons are still present in the danger zone.
- The drive shafts between motor and gear unit must be completely covered.
On completion of the work and once all persons have left the danger zone
- the access doors (D-B04-01) must be locked again and
- the keys deposited in the CP2 control room.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.3.2 Stand BD1

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D02-01 to D-D02-06 (and sliding lattice gate EB-D02-07)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of stand BD1</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>DR10</td><td colspan="4">- Roller table in area of descaler</td></tr>
<tr><td>DR20, DR30, DR40, DR50</td><td colspan="4">- Work and extension roller table upstream and downstream of stand BD1</td></tr>
<tr><td>DE10, DK10, DE20, DK20</td><td colspan="4">- Shifting device/gripper tilter upstream and downstream of stand BD1</td></tr>
<tr><td>DE15, DE25</td><td colspan="4">- Chain-type transfer upstream and downstream of stand BD1</td></tr>
<tr><td>DH00</td><td colspan="4">- Stand BD1</td></tr>
<tr><td>DN20</td><td colspan="4">- Roll changing cylinder</td></tr>
<tr><td></td><td colspan="4">- Hydraulic and pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-D02-01 to D-D02-06) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and when all persons have left the danger zone and the access doors (D02-01 to D02-06) have been closed again,
- the responsible person may activate the acknowledgement button which is arranged outside in the vicinity of the corresponding access door (D-D02-01 to D-D02-06).
- the doors to the relevant danger zone are then electrically locked again (with guard locking)by the machine's control system.
- the plant can be put back into operation.

Access to the danger zone (Danger zone 1) is protected by a sliding lattice grate (with safety switch with guard lock) in the area of the mill stands.
The request for opening the sliding lattice grate or for resetting it is made via a request or acknowledgement button which is arranged outside the danger zone.
Once work has been completed and when all persons have left the danger zone and the sliding lattice grate (EB-D02-07) has been closed again,
- the responsible person may activate the acknowledgement button which is arranged outside the danger zone in the vicinity of the sliding lattice grate (EB-D02-07).
- the sliding lattice grate is electrically interlocked (with guard locking) from the machine control system.
- the plant can be put back into operation.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.3.3 Stand BD1 (preparation for roll change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D03-01 to D-D03-05 (and sliding lattice gate EB-D02-07)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Stand changing equipment
(access via doors D-D03-01 to D-D03-05)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Setting up for mill stand change</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
DN10, DN20, DN30 - Stand changing equipment
- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="5"><table>
<tr><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="3">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
</table></td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept closed during machine movements.
Prior to entering the danger zone (Danger zone 2) the request button arranged in the vicinity of the respective door (D-D03-01 and D-D03-05) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
- the sliding lattice grate (EB-D02-07) is closed and electrically locked.
- the unit (stand changing equipment) has been taken out of operation.
- operation (stand changing equipment) has been transferred to the control station „Local Desk“ (LDBD)
.
Work can be carried out in the (Danger zone 2).

Once work has been completed and all persons have left the danger zone and the access doors (D-D03-01 to D-D03-05) have been closed again,
- the responsible person may activate the acknowledgement button arranged outside near the corresponding door (D-D03-01 to D-D03-05).
- the doors to the danger zone concerned are again electrically closed by the machine control system.
- the unit (stand changing equipment) can be put into operation again.

All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.3.4 Stand BD1 (maintenance (service) and repair, set-up/mill stand change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D03-01 to D-D03-05 (and sliding lattice gate EB-D02-07)</td></tr>
<tr><td>Access to:</td><td colspan="5">1. Danger zone 'Stand changing equipment'
(access via doors D-D03-01 to D-D03-05)
2. Danger zone 'Stand BD1'
(access via the sliding lattice grate EB-D02-07)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Setting up and mill stand change</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td colspan="2">Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td colspan="2">X</td><td></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>DN10, DN20, DN30</td><td colspan="4">1. Stand changing equipment at stand BD1
(with access via doors D-D03-01 to D-D03-05)</td></tr>
<tr><td></td><td colspan="4">2. Stand BD1
(with access via the sliding lattice grate EB-D02-07)</td></tr>
<tr><td>DH00</td><td colspan="4">- Millstand</td></tr>
<tr><td>DR20, DR30,
DR40, DR50</td><td colspan="4">- Work and extension roller tables
upstream and downstream of stand BD1</td></tr>
<tr><td>DE10, DK10,
DE20, DK20</td><td colspan="4">- Shifting device/gripper tilter
upstream and downstream of stand BD1</td></tr>
<tr><td>DE15, DE25</td><td colspan="4">- Chain-type transfer upstream and downstream of stand BD1</td></tr>
<tr><td></td><td colspan="4">3. Hydraulic and pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 2) the request button arranged in the vicinity of the respective door (D-D03-01 to D-D03-05) must be activated.
Condition:
- the sliding lattice grate (EB-D02-07) is closed and electrically locked.
- the Automatic mode (stand changing equipment) has been taken out of operation.
- operation of the stand changing equipment has been transferred to the "Local Desk" (LDBD1).
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 2).
To enter the danger zone (Danger zone 1) the request button placed in the vicinity of the sliding lattice grate must be actuated.
Condition:
- the Automatic mode (BD stand) has been taken out of operation.
- operation of the BD stand has been transferred to the "Local Desk" (LDBD1).
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and all persons have left the danger zone again, the responsible person may actuate the acknowledgement button placed outside of the danger zone in the area of the sliding lattice grate.
The BD stand can be restarted and the Automatic mode (BD stand) put into operation again.
Condition: The sliding lattice grate (EB-D02-07) is closed and electrically locked.
Once work has been completed and when all persons have left the danger zone and the access doors (D-D03-01 to D-D03-05) have been closed again,
- the responsible person may activate the acknowledgement button arranged outside near the corresponding door (D-D03-01 to D-D03-05).
- the doors to the relevant danger zone are then locked by the machine control system
.- the stand changing equipment can be switched on again and
the Automatic mode put into operation again.
<u>Description continued on following page:</u></td></tr>
</table>

	Description continued: **Special operating conditions**: For setting-up it may be required that persons stay in the danger zone (**Danger zone 2**). Prior to **entering** the danger zone (**Danger zone 2**) the request button arranged in the vicinity of the respective door (D-D03-01 to D-D03-05) must be activated. As soon as the access has been granted from the control station "**CP2**", the control commands are transferred to the local control desk "**Local Desk**" (**LDBD1**). Upon activation of the key-operated switch at the '**Local Desk**', all plant components in the danger zone (**stand changing equipment**) may only be moved in **Semi-automatic** or **Manual** mode from the **'Local desk'**. During this process, all movements are performed step by step and one after the other. All movements may only be carried once the operator has made sure that there are **no persons in the danger zone/range of movement**. All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person who stays outside the range of movement. The movements are indicated by an **optical** and **acoustic signal**. **Note**: **Automatic operation** of the stand changing equipment is only permitted when all persons have left the danger zone and the **doors** D-D03-01 to D-D03-05 have been closed and **locked**. All doors must be equipped with an escape unlocking mechanism on the inside. For detailed descriptions on opening and closing an electrically locked door, please refer to **Section 4.6.1**.

4.6.3.3.5 Stand BD1 (inspection under load)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D03-01 to D-D03-05 (and sliding lattice gate EB-D02-07)</td></tr>
<tr><td>Access to:</td><td colspan="5">1. Danger zone 'Stand changing equipment'
(access via doors D-D03-01 to D-D03-05)
2. Danger zone 'Stand BD1'
(access via the sliding lattice grate EB-D02-07)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Process monitoring, adjusting work; for Inspection under load</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td>YES</td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
DN10, DN20, DN30 - Stand changing equipment at stand BD1
- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="5"><table>
<tr><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
</table></td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 2 (stand changing equipment)) the request button arranged in the vicinity of the respective door (D-D03-01 to D-D03-05) must be activated.
The doors to the danger zone can/may only be opened and the danger zone (Danger zone 2 (stand changing equipment)) can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
- the sliding lattice grate (EB-D02-07) is closed and electrically locked.
- the Automatic mode (stand changing equipment) has been taken out of operation.
To enter the danger zone (Danger zone 1) the request button placed in the vicinity of the sliding lattice grate must be actuated.
Condition:
- the Automatic mode (BD stand) has been taken out of operation.
- Inspection under load has been selected,
i.e.
for Inspection under load, when the sliding lattice grate is unlocked and opened,
the rolling process at the BD stand may only be carried out at reduced speed (max. 3 m/s)
and in a sequence (Semi-automatic).
- Access to this danger zone is allowed only for authorised and appointed personnel.
- All persons who have entered or left the danger zone must sign in and out in the control room CP2 so that the plant operator is informed about as to whether persons are still present in the danger zone.
- All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person outside the range of movement.
Once the work/inspections have been completed and all persons have left the danger zone (Danger zone 1 (BD stand)) again, the in-charge person may activate the acknowledgement button outside of the danger zone in the area of the sliding lattice grate.
The BD stand can be restarted and the Automatic mode (BD stand) put into operation again.
Condition: The sliding lattice grate (EB-D02-07) is closed and electrically locked.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.3.6 Stand BD1 (access to gripper tilter service platform)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D05-01 or D-D05-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of stand BD1</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>DH00</td><td colspan="4">- Stand BD1</td></tr>
<tr><td>DK10, DK20</td><td colspan="4">- Gripper tilter upstream and downstream of stand BD1</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP2".
Before entering the danger zone, the relevant plant area must be switched off from the CP2 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-D05-01 or D-D05-02) must be locked again and
- the keys deposited in the CP2 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.3.7 Extension roller table and cross-transfer unit downstream of stand BD1

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-D06-01 to D-D06-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Extension roller table and cross-transfer unit downstream of stand BD1</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>DR40, DR50, DR60</td><td colspan="4">- Work and extension roller table upstream and downstream of stand BD1</td></tr>
<tr><td>DH50</td><td colspan="4">- Stand BD1 (main drive)</td></tr>
<tr><td>DC10</td><td colspan="4">- Cross transfer</td></tr>
<tr><td>ER10</td><td colspan="4">- Extension roller table upstream of stand BD2</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP2".
Before entering the danger zone, the relevant plant area must be switched off from the CP2 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-D06-01 to D-D06-03) must be locked again and
- the keys deposited in the CP2 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.4 Sub-section 4 (equipment in area of stand BD2)

4.6.3.4.1 Extension roller table upstream of stand BD2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-E01-01 and D-E01-02</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zone
Extension roller table (and cross transfer unit) upstream of stand BD2</td></tr>
<tr><td>Access required for:</td><td colspan="6">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>DC10</td><td colspan="5">- Cross transfer</td></tr>
<tr><td>ER10</td><td colspan="5">- Extension roller table upstream of stand BD2</td></tr>
<tr><td>ER20</td><td colspan="5">- Work roller table upstream of stand BD2</td></tr>
<tr><td>EH50</td><td colspan="5">- Stand BD2 (main drive)</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="3">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP3".
Before entering the danger zone, the relevant plant area must be switched off from the CP3 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-E01-01 and D-E01-02) must be locked again and
- the keys deposited in the CP3 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.4.2 Stand BD2 (drive)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E02-01 and D-E02-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of stand BD2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
Exception: "Troubleshooting"
EH50 - Stand BD2 (main drive)
- Hydraulic and pneumatic equipment</td></tr>
<tr><td rowspan="2">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td>X</td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP3".
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.

Condition:
- Access to this danger zone is allowed only for authorised and appointed personnel.

- All persons who have entered or left the danger zone must sign in and out in control room CP3 so that the plant operator is aware whether persons are still present in the danger zone. The drive shafts between motor and gear unit must be completely covered.
On completion of the work and once all persons have left the danger zone
- the access doors (D-E02-01 and D-E02-02) must be locked again and
- the keys deposited in the CP3 control room.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.4.3 Stand BD2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E03-01 to D-E03-03 (and sliding lattice grate EB-E03-04)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of stand BD2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>ER20, ER10,
ER30, ER40</td><td colspan="4">- Work and extension roller table upstream and downstream of stand BD2</td></tr>
<tr><td>EE10, EE20
EK10, EK20</td><td colspan="4">- Shifting device upstream and downstream of stand BD2</td></tr>
<tr><td>EE15, EE25</td><td colspan="4">- Chain-type transfer upstream and downstream of stand BD2</td></tr>
<tr><td>EH50</td><td colspan="4">- Stand BD2 (main drive)</td></tr>
<tr><td>EN20</td><td colspan="4">- Roll changing cylinder</td></tr>
<tr><td></td><td colspan="4">- Hydraulic and pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-E03-01 to D-E03-03) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and when all persons have left the danger zone and the access doors (D-E03-01 to D-E03-03) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-E03-01 to D-E03-03).
- the doors to the relevant danger zone are then electrically locked again (with guard locking) by the machine's control system.
- the plant can be put back into operation.

Access to the danger zone (Danger zone 1) is protected by a sliding lattice grate (with safety switch with guard lock) in the area of the mill stands.
The request for opening the sliding lattice grate or for resetting it is made via a request or acknowledgement button which is arranged outside the danger zone.
Once work has been completed and when all persons have left the danger zone and the sliding lattice grate (EB-E03-04) has been closed again,
- the in-charge person may activate the acknowledgement button arranged outside the danger zone near the sliding lattice grate (EB-E03-04).
- the sliding lattice grate is electrically interlocked (with guard locking) from the machine control system.
- the plant can be put back into operation.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.4.4 Stand BD2 (preparation for roll change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E04-01 to D-E04-04 (and sliding lattice grate EB-E03-04)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Stand changing equipment</td></tr>
<tr><td>Access required for:</td><td colspan="5">Setting up for mill stand change</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
EN10, EN20, EN30 - Stand changing equipment at stand BD2
- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept closed during machine movements.
Prior to entering the danger zone (Danger zone 2) the request button arranged in the vicinity of the respective door (D-E04-01 to D-E04-04) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
- The sliding lattice grate (EB-E03-04) is closed and electrically locked.
- The unit (stand changing equipment) has been taken out of operation.
- Operation (stand changing equipment) has been transferred to the "Local Desk" (LDBD)
.
Work can be carried out in the (Danger zone 2).

Once work has been completed and all persons have left the danger zone and the access doors (D-E04-01 to D-E04-04) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-E04-01 to D-E04-04).
- the doors to the danger zone concerned are again electrically closed by the machine control system.
- the unit (stand changing equipment) can be put into operation again.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.4.5 Stand BD 2 (maintenance (service) and repair, set-up/mill stand change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E04-01 to D-E04-04 (and sliding lattice grate EB-E03-04)</td></tr>
<tr><td>Access to:</td><td colspan="5">1. Danger zone 'Stand changing equipment'
(access via doors D-E04-01 to D-E04-04)
2. Danger zone 'Stand BD2'
(access via the sliding lattice grate EB-E03-04)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Setting up and mill stand change</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>EN10, EN20, EN30</td><td colspan="4">1. Stand changing equipment at stand BD2
(with access via doors D-E04-01 to D-E04-04)</td></tr>
<tr><td></td><td colspan="4">2. Stand BD2
(with access via the sliding lattice grate EB-E03-04)</td></tr>
<tr><td>EH00</td><td colspan="4">- Millstand</td></tr>
<tr><td>ER20, ER10,
ER30, ER40</td><td colspan="4">- Work and extension roller tables
upstream and downstream of stand BD2</td></tr>
<tr><td>EE10, EE20
EK10, EK20</td><td colspan="4">- Shifting device/gripper tilter
upstream and downstream of stand BD2</td></tr>
<tr><td>EE15, EE25</td><td colspan="4">- Chain-type transfer upstream and downstream of stand BD2</td></tr>
<tr><td></td><td colspan="4">3. Hydraulic and pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 2) the request button arranged in the vicinity of the respective door (D-E04-01 to D-E04-04) must be activated.
Condition:
- The sliding lattice grate (EB-E03-04) is closed and electrically locked.
- The Automatic mode (stand changing equipment) has been taken out of operation.
- Operation of the stand changing equipment has been transferred to the "Local Desk" (LDBD2).
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 2).
To enter the danger zone (Danger zone 1) the request button placed in the vicinity of the sliding lattice grate must be actuated.
Condition:
- the Automatic mode (BD stand) has been taken out of operation.
- operation of the BD stand has been transferred to the "Local Desk" (LDBD2) .
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and all persons have left the danger zone again, the responsible person may actuate the acknowledgement button placed outside of the danger zone in the area of the sliding lattice grate.
The BD stand can be restarted and the Automatic mode (BD stand) put into operation again.
Condition: The sliding lattice grate (EB-E03-04) is closed and electrically locked.
Once work has been completed and when all persons have left the danger zone and the access doors (D-E04-01 to D-E04-04) have been closed again,
- the responsible person may activate the acknowledgement button arranged outside near the corresponding door (D-E04-01 to D-E04-04).
- the doors to the relevant danger zone are then locked by the machine control system
- the stand changing equipment can be switched on again and the Automatic mode put into operation again.
<u>Description continued on following page:</u></td></tr>
</table>

	Description continued: **Special operating conditions**: For setting-up it may be required that persons stay in the danger zone (**Danger zone 2**). Prior to **entering** the danger zone (**Danger zone 2**) the request button arranged in the vicinity of the respective door (D-E04-01 to D-E04-04) must be activated. As soon as access has been granted from the **CP3** control room, the control commands are transferred to the '**Local Desk**' (**LDBD2**). Upon activation of the key-operated switch at the '**Local Desk**', all plant components in the danger zone (**stand changing equipment**) may only be moved in **Semi-automatic** or **Manual** mode from the **'Local desk'**. During this process, all movements are performed step by step and one after the other. All movements may only be carried once the operator has made sure that there are **no persons in the danger zone/range of movement**. All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person who stays outside the range of movement. The movements are indicated by an **optical** and **acoustic signal**. **Note**: **Automatic operation** of the stand changing equipment is only permitted when all persons have left the danger zone and the **doors** D-E04-01 to D-E04-04 have been closed and **locked**. All doors must be equipped with an escape unlocking mechanism on the inside. For detailed descriptions on opening and closing an electrically locked door, please refer to **Section 4.6.1**.

4.6.3.4.6 Stand BD2 (inspection under load)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E04-01 to D-E04-04 (and sliding lattice grate EB-E03-04)</td></tr>
<tr><td>Access to:</td><td colspan="5">1. Danger zone 'Stand changing equipment'
(access via doors D-E04-01 to D-E04-04)
2. Danger zone 'Stand BD2'
(access via the sliding lattice grate EB-E03-04)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Process monitoring, adjusting work; for Inspection under load</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td>YES</td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
EN10, EN20, EN30 - Stand changing equipment at stand BD2
- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="5"><table>
<tr><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="3">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
</table></td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 2 (stand changing equipment)), the request button arranged in the vicinity of the respective door (D-E04-01 to D-E04-04) must be activated.
The doors to the danger zone can/may only be opened and the danger zone (Danger zone 2 (stand changing equipment)) can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
- the sliding lattice grate (EB-E03-04) is closed and electrically locked.
- the Automatic mode (stand changing equipment) has been taken out of operation.
To enter the danger zone (Danger zone 1) the request button in the vicinity of the light curtain must be activated.
Condition:
- the Automatic mode (BD stand) has been taken out of operation.
- inspection under load has been selected,
i.e.
for Inspection under load, when the sliding lattice grate is unlocked and opened,
the rolling process at the BD stand may only be carried out at reduced speed (max. 3 m/s) and in a sequence (Semi-automatic).
- Access to this danger zone is allowed only for authorised and appointed personnel.
- All persons who have entered or left the danger zone must sign in and out in the control room CP3 so that the plant operator is aware whether persons are still present in the danger zone.
- All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person outside the range of movement.
Once the work/inspections have been completed and all persons have left the danger zone (Danger zone 1 (BD stand)) again, the in-charge person may activate the acknowledgement button outside of the danger zone in the area of the light curtain.
The BD stand can be restarted and the Automatic mode (BD stand) put into operation again.
Condition: The sliding lattice grate (EB-E03-04) is closed and electrically locked.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.4.7 Extension roller table/cross-transfer unit (and switch) downstream of stand BD2

Safety element (doors/light curtain):	**D-E05-01 and D-E05-02**			
Access to:	Danger zone **Extension roller table/cross-transfer unit (and switch) downstream of stand BD2**			
Access required for:	-			
during:	Maintenance (service)/ Repair work	Troubleshooting/ Fault elimination	Use/Operation (workplace)	Setting-up (workplace)
	X	**X**		
Access conditions:	Drives/hydraulic and pneumatic movements safely switched off (The **applicable version of the motors and components list** must be observed):			
	ER30, ER40	- Work and extension roller table downstream of stand BD2		
	EC20	- Cross transfer		
	EH50	- Stand BD2 (main drive)		
		- Hydraulic/pneumatic equipment		

Design of the safety element (interlocking):	Access door/sliding lattice grate			Light curtain	
	Mechanically **locked**	Electrically		with **request/ acknowledgement button**	designed acc to: PLr (EN ISO 13849-1)
		locked	with **request/ acknowledgement button**		
	X				**-**

Design:	The danger zone must be kept **locked** during machine movements. The user must assign the keys to the designated persons. The keys for the access doors to the danger zone are deposited in the control room "**CP3**". Before entering the danger zone, the relevant **plant area must be switched off** from the **CP3** control room. The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once **all hazardous movements** in the danger zone have stopped. Work can be carried out in the danger zone (**Danger zone 1**). Access to this danger zone is permitted only for **authorised and designated personnel**. On completion of the work and once **all persons** have left the danger zone - the access doors (D-E05-01 and D-E05-02) must be **locked** again and - the keys deposited in the **CP3** control room. The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again. All doors must be equipped with an escape unlocking mechanism on the inside.

4.6.3.4.8 Extension roller table and cross-transfer unit downstream of stand BD2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E06-01 and D-E06-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Extension roller table/cross-transfer unit (and switch) downstream of stand BD2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>ER30, ER40</td><td colspan="5">- Work and extension roller table downstream of stand BD2</td></tr>
<tr><td>EH50</td><td colspan="5">- Stand BD2 (main drive)</td></tr>
<tr><td>EC20</td><td colspan="5">- Cross transfer</td></tr>
<tr><td>GR10</td><td colspan="5">- Extension roller table upstream of tandem mill stand</td></tr>
<tr><td>GR20</td><td colspan="5">- Liftable and lowerable work roller table upstream of tandem mill stand</td></tr>
<tr><td>GG50, GH50, GG150</td><td colspan="5">- Mill stands UR, E and UF (main drive)</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP3".
Before entering the danger zone, the relevant plant area must be switched off from the CP3 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-E06-01 and D-E06-02) must be locked again and
- the keys deposited in the CP3 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.4.9 Hot/crop saw 1

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="4">D-E07-01 and D-E07-02</td></tr>
<tr><td>Access to:</td><td colspan="4">Danger zone
Hot/crop saw 1</td></tr>
<tr><td>Access required for:</td><td colspan="4">Changing the saw blade</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="4">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
EH50 - Stand BD2 (main drive)
ER40 - Extension roller table downstream of stand BD2
ES10 - Crop saw with high-pressure water station
- Hydraulics/pneumatics
(Pressurised systems (also with machine switched off))</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="4"><table>
<tr><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="3">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
</table></td></tr>
<tr><td>Design:</td><td colspan="4">The danger zones must be kept closed during machine movement.
Before entering the danger zone (Danger zones 1 and 2):
- the request button located in the vicinity of the corresponding access door (D-E07-01 and D-E07-02) must be activated.
- the control commands for the crop saw must have been transferred from the CP4 control room to the 'Local Desk' (LDHS1),
- the crop saw as well as the extension roller tables downstream of stand BD2 must have been taken out of operation.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1/2).
Before working on the high-pressure water station
- The supply valve at the pump station (high-pressure water station) must be closed and secured against restarting by means of a lock.
- A prohibition sign must be affixed at the supply valve of the pump station (high-pressure water station): 'Do not switch on'.
Once work has been completed and when all persons have left the danger zone and the access doors have been closed again,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door.
- the control commands for the saw are transferred again from the 'Local Desk' (LDHS1) to the control room 'CP4'.
- the doors to the relevant danger zone are once again electrically locked by the machine control system.
- the plant can be put into operation again.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.4.10 Hot/crop saw 1 (chip container cellar)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-E08-01</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zones
Hot/crop saw 1 (chip container cellar)</td></tr>
<tr><td>Access required for:</td><td colspan="6">Access to chip container cellar</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td colspan="2">Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td colspan="2"></td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>ES10</td><td colspan="5">- The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- High-pressure water station</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td>Mechanically locked</td><td colspan="2">Electrically</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td><td></td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.

Condition:
- Access to this danger zone is allowed only for authorised and appointed personnel.

- The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- The high-pressure water station is depressurized.
On completion of work and once all persons have left the danger zone again, the access doors must be reclosed and the keys deposited in the control room "CP4".

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.4.11 Hot/crop saw 1 (scrap container cellar)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-E09-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Hot/crop saw 1 (scrap container cellar)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Access to scrap container cellar</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td colspan="2">YES</td><td></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>ER40</td><td colspan="5">- Extension roller table downstream of stand BD2</td></tr>
<tr><td>ES10</td><td colspan="5">- Crop saw with high-pressure water station</td></tr>
<tr><td>EQ10</td><td colspan="5">- Scrap chute/flap</td></tr>
<tr><td></td><td colspan="5">- Hydraulics/pneumatics
(Pressurised systems (also with machine switched off))</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-E09-01) must be activated.
Condition:
- The required access doors to the danger zones must be designed such that they can only be opened when the flaps on the chutes of the scrap container are closed.
- Conversely, the flaps on the chutes of the scrap container can only be opened when the access doors to the danger zones are closed and locked (access control system with safety switch and guard locking).
- The saw is in an operationally secure position (the saw is retracted and is not cutting).
- The high-pressure water station is depressurized.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and when all persons have left the danger zone and the access door (D-E09-01) has been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-E09-01).
- the doors to the relevant danger zone are then electrically locked again (with guard locking)by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.5 Sub-section 5 (equipment in the area of the tandem mill stand)

4.6.3.5.1 Extension roller table (and cross-transfer unit) upstream of tandem mill stand

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G01-01 and D-G01-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Extension roller table (and cross-transfer unit) upstream of tandem mill stand</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>GR10</td><td colspan="4">- Extension roller table upstream of tandem mill stand</td></tr>
<tr><td>GR20</td><td colspan="4">- Liftable and lowerable work roller table upstream of tandem mill stand</td></tr>
<tr><td>EC20</td><td colspan="4">- Cross transfer</td></tr>
<tr><td>GG50, GH50, GG150</td><td colspan="4">- Mill stands UR, E and UF (main drive)</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
Before entering the danger zone, the relevant plant area must be switched off from the CP4 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-G01-01 and D-G01-02) must be locked again and
- the keys deposited in the CP4 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.5.2 Tandem mill stand (drive)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G05-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of tandem mill stand</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
Exception: „Troubleshooting"</td></tr>
<tr><td></td><td>GG50, GH50, GG150</td><td colspan="4">- Mill stands UR, E and UF (main drive)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.

Condition:
- Access to this danger zone is allowed only for authorised and appointed personnel.

- All persons who have entered or left the danger zone must sign in and out in the control room CP2 so that the plant operator is aware whether persons are still present in the danger zone.
- The drive shafts between motor and gear unit must be completely covered.
On completion of the work and once all persons have left the danger zone
- the access doors (D-G05-01) must be locked again and
- the keys deposited in the CP4 control room.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.5.3 Tandem mill stand

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G03-01 to D-G03-04 (and light curtain (LB-G03-05)
(or other access control system (access door/sliding lattice grate (D-/EB-G03-05))))</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of tandem mill stand</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>GR10, GR40</td><td colspan="4">- Extension roller tables upstream and downstream of tandem mill stand</td></tr>
<tr><td>GR20, GR30</td><td colspan="4">- Liftable and lowerable work roller table upstream and downstream of tandem mill stand</td></tr>
<tr><td>GE10, GE20</td><td colspan="4">- Shifting and centering device up and downstream of tandem stand</td></tr>
<tr><td>GG10, GH10, GG110</td><td colspan="4">- Mill stands UR, E and UF</td></tr>
<tr><td>GN20</td><td colspan="4">- Stand shifting cylinder</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="3">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-G03-01 to D-G03-04) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and when all persons have left the danger zone and the access doors (D-G03-01 to D-G03-04) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-G03-01 to D-G03-04).
- the doors to the relevant danger zone are then electrically locked again (with guard locking)by the machine's control system.
- the plant can be put back into operation.

Access to the danger zone (Danger zone 1) is protected in the area of the mill stands with a light curtain or other access control systems (door or sliding lattice grate (with a safety switch with guard lock)).
The request to deactivate and activate the light curtain (LB-G03-05) or to open the door or sliding lattice grate (D- or EB-G03-05) or reset it is given via a request or acknowledgment button located outside the danger zone.
Once work has been completed and all persons have left the danger zone again
- the access door/sliding lattice grate (D- or EB-G03-05) must be locked again,
- the in-charge person may activate the acknowledgement button arranged outside the danger zone near the access door/sliding lattice grate (D- or EB-G03-05).
- the access door/sliding lattice grate (D- or EB-G03-05) is electrically locked again (with guard locking) from the machine control system
or
- the in-charge person may activate the acknowledgement button arranged outside the danger zone in the area of the light curtain (LB-G03-05).
- the light curtain (LB-G03-05) is activated again.
The plant can then be put into operation again.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to Sections 4.6.1 and 4.6.2.</td></tr>
</table>

4.6.3.5.4 Tandem mill stand (preparation for roll change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G04-01 to D-G04-03 (and light curtain (LB-G03-05)
(or other access control system (access door/sliding lattice grate (D-/EB-G03-05))))</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Stand changing equipment</td></tr>
<tr><td>Access required for:</td><td colspan="5">Setting-up for mill stand changing</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
GN10, GN20, GN30 - Stand changing equipment at tandem mill stand
- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept closed during machine movements.
Prior to entering the danger zone (Danger zone 2), the request button located in the vicinity of the corresponding door (D-G04-01 to D-G04-03), must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
-The light curtain LB-G03-05 is activated.
or
- The access door/sliding lattice grate (D- or EB-G03-05) is electrically locked (with guard locking).
- The unit (stand changing equipment) has been taken out of operation.
- Operation (stand changing equipment) has been transferred to the "Local Desk" (LDTM1)
.
Work can be carried out in the danger zone (Danger zone 2).

Once work has been completed and all persons have left the danger zone and the access doors (D-G04-01 to D-G04-03) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-G04-01 to D-G04-03).
- the doors to the danger zone concerned are again electrically closed by the machine control system.
- the unit (stand changing equipment) can be put into operation again.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to Sections 4.6.1 and 4.6.2.</td></tr>
</table>

4.6.3.5.5 Tandem mill stand (maintenance (service)/repair, set-up/mill stand change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="4">D-G04-01 to D-G04-03 (and light curtain (LB-G03-05)
(or other access control system (access door/sliding lattice grate (D-/EB-G03-05))))</td></tr>
<tr><td>Access to:</td><td colspan="4">1. Danger zone 'Stand changing equipment'
(access via doors D-G04-01 to D-G04-03)
2. Danger zone 'Tandem mil stand'
(access via light curtain LB-G03-05 or access door/sliding lattice grate D-/EB-G03-05)</td></tr>
<tr><td>Access required for:</td><td colspan="4">Set-up work and mill stand changing</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="4">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
GN10, GN20, GN30 — 1. Stand changing equipment at tandem mill stand (with access via doors D-G04-01 to D-G04-03)
2. Tandem mill stand (with access via light curtain LB-G03-05 or access door/sliding lattice grate D-/EB-G03-05)
GG10, GH10, GG110 — - Mill stands UR, E and UF
GR20, GR10 GR30, GR40 — - Liftable and lowerable work and extension roller tables upstream and downstream of tandem mill stand
GE10, GE20 — - Shifting and centering device upstream and downstream of tandem mill stand
3. Hydraulic and pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="4">
<table>
<tr><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="3">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
</table>
</td></tr>
<tr><td>Design:</td><td colspan="4">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 2) the request button arranged in the vicinity of the respective door (D-G04-01 to D-G04-03) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
-The light curtain LB-G03-05 is activated.
or
- The access door/sliding lattice grate (D- or EB-G03-05) is electrically locked (with guard locking).
- The unit (stand changing equipment) has been taken out of operation.
- Operation (stand changing equipment) has been transferred to the "Local Desk" (LDTM1)
.
Work can be carried out in the danger zone (Danger zone 2).
To enter the danger zone (Danger zone 1) the request button arranged in the vicinity of the light curtain LB-G03-05 or at the access door/sliding lattice grate (D- or EB-G03-05) must be activated.
Condition:
- the Automatic mode (tandem mill stands) has been taken out of operation.
- Operation of the tandem mill stands has been transferred to the "Local Desk" LDTM1.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and once all persons have the danger zone again, the in-charge person may activate the acknowledgement button located outside the danger zone in the area of the light curtain LB-G03-05 .
The tandem mill stands can be restarted and the Automatic mode (tandem mill stands) put into operation again.
Condition: The light curtain LB-G03-05 is activated and
the access door/sliding lattice grate (D- or EB-G03-05) is electrically locked.

<u>Description continued on following page:</u></td></tr>
</table>

Description continued:

Once work has been completed and when **all persons** have left the danger zone and the access doors (D-G04-01 to D-G04-03) have been closed again,

- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-G04-01 to D-G04-03).
- the doors to the relevant danger zone are then locked by the machine control system
- the **stand changing equipment** can be switched on again and the **Automatic mode put into operation** again.

Special operating conditions:
For setting-up it may be required that persons stay in the danger zone (**Danger zone 2**).

Prior to **entering** the danger zone (**Danger zone 2**) the request button arranged in the vicinity of the respective door (D-G04-01 to D-G04-03) must be activated.

As soon as access has been granted from the **CP4** control room, the control commands are transferred to the '**Local Desk**' (**LDTM1**). Upon activation of the key-operated switch at the '**Local Desk**', all plant components in the danger zone (**stand changing equipment**) may only be moved in **Semi-automatic** or **Manual** mode from the **local control desk**. During this process, all movements are performed step by step and one after the other.

All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person who stays outside the range of movement.

The movements are indicated by an **optical** and **acoustic signal**.

Note: **Automatic operation** of the stand changing equipment is only permitted when all persons have left the danger zone and the **doors** D-G04-01 to D-G04-03 have been closed and **locked**.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to **Sections 4.6.1 and 4.6.2**.

4.6.3.5.6 Tandem mill stand (inspection under load)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G04-01 to D-G04-03 (and light curtain (LB-G03-05)
(or other access control system (access door/sliding lattice grate (D-/EB-G03-05))))</td></tr>
<tr><td>Access to:</td><td colspan="5">1. Danger zone 'Stand changing equipment'
(access via doors D-G04-01 to D-G04-03)
2. Danger zone 'Tandem mil stand'
(access via light curtain LB-G03-05 or access door/sliding lattice grate D-/EB-G03-05)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Process monitoring, for inspection under load</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td>YES</td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
GN10, GN20, GN30 - Stand changing equipment at tandem mill stand
- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="5"><table>
<tr><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
</table></td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the danger zone (Danger zone 2 (stand changing equipment)), the request button arranged near the corresponding door (D-G04-01 to D-G04-03) must be activated.
The doors to the danger zone can/may only be opened and the danger zone (Danger zone 2 (stand changing equipment)) can/may only be entered once all hazardous movements in the danger zone have stopped.
Condition:
-The light curtain LB-G03-05 is activated.
or
- The access door/sliding lattice grate (D- or EB-G03-05) is electrically locked (with guard locking).
- The unit (stand changing equipment) has been taken out of operation.
To enter the danger zone (Danger zone 1 (tandem mill stands)) to monitor the process, the request button located near the light curtainLB-G03-05 or at the access door/sliding lattice grate (D- or EB-G03-05) must be activated.
Condition:
- Inspection under load has been selected, i.e.
with Inspection under load, when the light curtain is deactivated (LB-G03-05)
or the access door/sliding lattice grate is unlocked and opened (D-/EB-G03-05)
the material flow at the tandem mill stand may only take place at reduced speed (max. 3 m/s) and in a sequence .
- Access to this danger zone is allowed only for authorised and appointed personnel.
- All persons who have entered or left the danger zone must sign in and out in the control room CP4
so that the plant operator is informed about as to whether persons are still present in the danger zone.
- All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person outside the range of movement.
On completion of the work/inspection and once all persons have left the danger zone (Danger zone 1 (tandem mill stands)), the person in charge may activate the acknowledgement button located outside the danger zone in the area of the light curtain (LB-G03-05) or on the access door/sliding lattice grate (D- or EB-G03-05).
The tandem mill stands can then be switched back to Automatic mode.
Condition: The light curtain (LB-G03-05) is activated and
the access door/sliding lattice grate (D- or EB-G03-05) is electrically locked.
The stand changing equipment can/may only be brought back into service
- once all persons have left the danger zone (danger zone 2),
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-G04-01 to D-G04-03) and
- the access doors (D-G04-01 to D-G04-03) are closed again and electrically locked.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to Sections 4.6.1 and 4.6.2.</td></tr>
</table>

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G06-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Extension roller table downstream of tandem mill stand</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>GG10, GH10, GG110</td><td colspan="4">- Mill stands UR, E and UF</td></tr>
<tr><td>GR30</td><td colspan="4">- Liftable and lowerable work roller table downstream of tandem mill stand</td></tr>
<tr><td>GR40</td><td colspan="4">- Extension roller table downstream of tandem mill stand</td></tr>
<tr><td>GZ20</td><td colspan="4">- Rail stamping machine</td></tr>
<tr><td>GZ30</td><td colspan="4">- Profile gauge</td></tr>
<tr><td>GC10, GK10</td><td colspan="4">- Cross-transfer unit with rail turning device downstream of tandem mill stand</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
Before entering the danger zone, the relevant plant area must be switched off from the CP4 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-G06-01) must be locked again and
- the keys deposited in the CP4 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.5.7 Rail stamping machine

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-G07-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Rail stamping machine</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td colspan="2"></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>GG50, GH50, GG150</td><td colspan="4">- Mill stands UR, E and UF (main drive)</td></tr>
<tr><td></td><td>GR30</td><td colspan="4">- Liftable and lowerable work roller table downstream of tandem mill stand</td></tr>
<tr><td></td><td>GR40</td><td colspan="4">- Extension roller table downstream of tandem mill stand</td></tr>
<tr><td></td><td>GZ20</td><td colspan="4">- Rail stamping machine</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-G07-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Condition:
- the Automatic mode has been taken out of operation.
- Operation has been transferred to the Local Desk 'LD Stamp'.

Once work has been completed and when all persons have left the danger zone and the access door (D-G07-01) has been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-G07-01).
- the doors to the relevant danger zone are then electrically locked again (with guard locking) by the machine's control system.
- the plant can be put back into operation.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.5.8 Extension roller table and cross-transfer unit (with rail turning device) downstream of tandem mill stand

Safety element (doors/light curtain):	**D-G08-01 to D-G08-03**			
Access to:	Danger zone **Extension roller table and cross-transfer unit (with rail turning device) downstream of tandem mill stand**			
Access required for:	-			
during:	Maintenance (service)/ Repair work	Troubleshooting/ Fault elimination	Use/Operation (workplace)	Setting-up (workplace)
	X	**X**		
Access conditions:	Drives/hydraulic and pneumatic movements safely switched off (The **applicable version of the motors and components list** must be observed):			
	GG50, GH50, GG150	- Mill stands UR, E and UF (main drive)		
	GR30	- Liftable and lowerable work roller table downstream of tandem mill stand		
	GR40	- Extension roller table downstream of tandem mill stand		
	GC10	- Cross-transfer unit downstream of tandem mill stand		
	JR10	- Feed roller table to the cooling bed/induction heating system		
		- Hydraulic/pneumatic equipment		

Design of the safety element (interlocking):	Access door/sliding lattice grate			Light curtain	
	Mechanically **locked**	Electrically		with **request/ acknowledgement button**	designed acc to: PLr (EN ISO 13849-1)
		locked	with **request/ acknowledgement button**		
	X				**-**

Design:	The danger zone must be kept **locked** during machine movements. The user must assign the keys to the designated persons. The keys for the access doors to the danger zone are deposited in the control room "**CP4**". Before entering the danger zone, the relevant **plant area must be switched off** from the **CP4** control room. The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once **all hazardous movements** in the danger zone have stopped. Work can be carried out in the danger zone (**Danger zone 1**). Access to this danger zone is permitted only for **authorised and designated personnel**. On completion of the work and once **all persons** have left the danger zone - the access doors (D-G08-01 to D-G08-03) must be **locked** again and - the keys deposited in the **CP4** control room. The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again. All doors must be equipped with an escape unlocking mechanism on the inside.

4.6.3.6 Sub-section 6 (equipment in the area of the cooling bed)

4.6.3.6.1 Profile gauge

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J01-01 (and sliding door SD-J01-02)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Profile gauge</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td colspan="2"></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>GG50, GH50, GG150</td><td colspan="4">- Mill stands UR, E and UF (main drive)</td></tr>
<tr><td></td><td>GR30</td><td colspan="4">- Liftable and lowerable work roller table downstream of tandem mill stand</td></tr>
<tr><td></td><td>GR40</td><td colspan="4">- Extension roller table downstream of tandem mill stand</td></tr>
<tr><td></td><td>GZ30</td><td colspan="4">- Profile gauge</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td></td><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-J01-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once
- all hazardous movements in the danger zone have stopped
- the profile gauge has been taken out of service
- the profile gauge is still in the mill line or the sliding door (SD-J01-02) is closed.
To remove the profile gauge from the mill line
- the mill stands UR, E und UF (main drive)
- the liftable and lowerable work roller table downstream of the tandem mill stand
- the extension roller table downstream of the tandem mill stand
- the profile gauge
must be taken out of service.
Operation of the profile gauge has been transferred to the 'Local Desk' (LD Gauge).
To bring the mill (also without profile gauge) back into service, the sliding door (SD-J01-02) must be closed and the acknowledgement button, located near the sliding door, activated.
The sliding door is electrically locked by the machine control system.
Before opening the sliding door (SD-J01-02) the request button located in the vicinity of the sliding door must be activated.
The sliding door can/may only be opened and the danger zone of the mill can/may only be entered once all hazardous movements in the danger zone of the mill
- Mill stands UR, E and UF (main drive)
- liftable and lowerable work roller table downstream of the tandem mill stand
- Extension roller table downstream of the tandem mill stand have stopped.
Once work has been completed and when all persons have left the danger zone (profile gauge) and the access door (D-J01-01) has been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the orresponding access door (D-J01-01).
- the doors to the relevant danger zone are then electrically locked again (with guard locking) by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For details on opening and closing an electrically locked door, see Section 4.6.1.</td></tr>
</table>

4.6.3.6.2 Hot/pendulum saw 2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J02-01 and D-J02-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Hot/pendulum saw 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">Changing the saw blade</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>GC10, GK10</td><td colspan="4">- Cross-transfer unit with rail turning device downstream of tandem mill stand</td></tr>
<tr><td></td><td>JR10</td><td colspan="4">- Feed roller table to the cooling bed/induction heating system</td></tr>
<tr><td></td><td>JS10</td><td colspan="4">- Hot/pendulum saw 2 with high-pressure water station</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td></td><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-J02-01 and D-J02-02) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the hot saw:
- the control commands for the hot saw must have been transferred from the 'CP4' control room to the Local Desk (LDHS2),
- the hot saw as well as the feed roller table for the cooling bed must be taken out of service
For work on the high-pressure water station the
- the supply valve at the pump station (high-pressure water station) must also be closed and secured against restarting by means of a lock.
- a prohibition sign must be affixed at the supply valve of the pump station (high-pressure water station): 'Do not switch on'.

Once work has been completed and when all persons have left the danger zone and the access doors have been closed again,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door.
- the control commands for the saw are transferred again from the Local Desk (LDHS2) to the CP4 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.6.3 Hot/pendulum saw 2 (chip container cellar)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J03-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Hot/pendulum saw 2 (chip container cellar)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Access to chip container cellar</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JS10</td><td colspan="4">- The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- High-pressure water station</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.

Condition:
- Access to this danger zone is allowed only for authorised and appointed personnel.
- The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- The high-pressure water station is depressurized.
Once the work has been completed and all persons have left the danger zone, the access doors must again be locked.
If necessary, the keys must again be deposited in the CP4 control room.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.6.4 Hot/pendulum saw 2 (scrap container cellar)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J04-01 and J04-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Hot/pendulum saw 2
(scrap container cellar with crop-end removal system and scrap container transfer car)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Emptying of the crop end and scrap container</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JS10</td><td colspan="4">- Hot/pendulum saw 2
The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- High-pressure water station</td></tr>
<tr><td></td><td>JQ10</td><td colspan="4">- Crop-end removal system</td></tr>
<tr><td></td><td>JQ30</td><td colspan="4">- Scrap container transfer car</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-J04-01 and D-J04-02) must be activated.
Condition:
- All hazardous movements (scrap container transfer car) in the danger zone have stopped.
- The required access doors to the danger zones must be designed such that they can only be opened when the flaps on the chutes of the scrap container are closed.
- Conversely, the flaps on the chutes of the scrap container can only be opened when the access doors to the danger zones are closed and locked (access control system with safety switch and guard locking).
- The saw is in an operationally secure position (the saw is retracted and is not cutting).
- The high-pressure water station is switched off.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and when all persons have left the danger zone and the access doors (D-J04-01 and J04-02) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-J04-01 and J04-02).
- the doors to the relevant danger zone are then electrically locked again (with guard locking)by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.6.5 Feed roller table (and cross-transfer unit) for the cooling bed/induction heating unit

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J05-01 to D-J05-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Feed roller table (and cross-transfer unit) for the cooling bed/induction heating unit</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>GC10, GK10</td><td colspan="4">- Cross-transfer unit with rail turning device downstream of tandem mill stand</td></tr>
<tr><td></td><td>JR10</td><td colspan="4">- Feed roller table to the cooling bed/induction heating system</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td></td><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td><td></td></tr>
<tr><td></td><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
Before entering the danger zone, the relevant plant area must be switched off from the CP4 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-J05-01 to D-J05-03) must be locked again and
- the keys deposited in the CP4 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.6.6 Hot/pendulum saw 3

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="4">D-J06-01 and D-J06-02</td></tr>
<tr><td>Access to:</td><td colspan="4">Danger zones
Hot/pendulum saw 3</td></tr>
<tr><td>Access required for:</td><td colspan="4">Changing the saw blade</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="4">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>GC10, GK10</td><td colspan="3">- Cross-transfer unit with rail turning device downstream of tandem mill stand</td></tr>
<tr><td>JR10</td><td colspan="3">- Feed roller table to the cooling bed/induction heating system</td></tr>
<tr><td>JS110</td><td colspan="3">- Hot/pendulum saw 3 with high-pressure water station</td></tr>
<tr><td>JQ120</td><td colspan="3">- Pusher</td></tr>
<tr><td>JT10 to JT50</td><td colspan="3">- Pinch roll unit induction heating system and cooling line</td></tr>
<tr><td></td><td colspan="3">- Hydraulics/pneumatics
(Pressurised systems (also with machine switched off))</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="4">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-J06-01 and D-J06-02) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the hot saw:
- the control commands for the hot saw must have been transferred from the 'CP4' control room to the Local Desk (LDHS3),
- the hot saw as well as the feed roller table for the cooling bed must be taken out of service
For work on the high-pressure water station the
- the supply valve at the pump station (high-pressure water station) must also be closed and secured against restarting by means of a lock.
- a prohibition sign must be affixed at the supply valve of the pump station (high-pressure water station): 'Do not switch on'.

Once work has been completed and when all persons have left the danger zone and the access doors have been closed again,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door.
- the control commands for the saw are transferred again from the Local Desk (LDHS3) to the CP4 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.

All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.6.7 Hot/pendulum saw 3 (chip container cellar)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-J07-01</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zones
Hot/pendulum saw 3
(chip container cellar)</td></tr>
<tr><td>Access required for:</td><td colspan="6">Emptying of the crop end and scrap container</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td colspan="2">Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td colspan="2"></td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JS110</td><td colspan="5">- Hot saw
The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- High-pressure water station</td></tr>
<tr><td></td><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td>Mechanically locked</td><td colspan="2">Electrically</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td><td></td><td></td></tr>
<tr><td>X</td><td></td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP4".
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.

Condition:
- Access to this danger zone is allowed only for authorised and appointed personnel.
- The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- The high-pressure water station is switched off.
Once the work has been completed and all persons have left the danger zone, the access doors must again be locked.
If necessary, the keys must again be deposited in the CP4 control room.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.6.8 Hot/pendulum saw 3 (scrap container cellar)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J08-01 and D-J08-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Hot/pendulum saw 3
(scrap container cellar with
crop-end removal system and
scrap container transfer car)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Emptying of the crop end and scrap container</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JS110</td><td colspan="4">- Hot/pendulum saw 3
The saw is in an operationally secure position
(the saw is retracted and is not cutting).
- High-pressure water station</td></tr>
<tr><td></td><td>JQ110</td><td colspan="4">- Crop-end removal system</td></tr>
<tr><td></td><td>JQ130</td><td colspan="4">- Scrap container transfer car</td></tr>
<tr><td></td><td>JQ120</td><td colspan="4">- Pusher</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-J08-01 and D-J08-02) must be activated.
Condition:
- All hazardous movements (scrap container transfer car) in the danger zone have stopped.
- The required access doors to the danger zones must be designed such that they can only be opened when the flaps on the chutes of the scrap container are closed.
- Conversely, the flaps on the chutes of the scrap container can only be opened when the access doors to the danger zones are closed and locked (access control system with safety switch and guard locking).
- The saw is in an operationally secure position (the saw is retracted and is not cutting).
- The high-pressure water station is switched off.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and when all persons have left the danger zone and the access doors (D-J08-01 and J08-02) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-J08-01 and J08-02).
- the doors to the relevant danger zone are then electrically locked again (with guard locking) by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.6.9 Rail head quenching and tempering area (upstream of induction heating unit)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J09-01 and D-J09-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Rail head quenching and tempering area
(upstream of induction heating unit)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JR10</td><td colspan="4">- Feed roller table to the cooling bed/induction heating system</td></tr>
<tr><td></td><td>JT10 to JT50</td><td colspan="4">- Pinch roll unit induction heating system and cooling line</td></tr>
<tr><td></td><td>JZ26</td><td colspan="4">- Induction heating unit (guide)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td></td><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP5.1".
Before entering the danger zone, the relevant plant area must be switched off from the CP5.1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-J09-01 and D-J09-02) must be locked again and
- the keys deposited in the CP5.1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.6.10 Rail head quenching and tempering area (induction heating unit)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J10-01 to D-J10-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Rail head quenching and tempering area
(induction heating unit)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Servicing/
Maintenance</td><td colspan="2">Troubleshooting/
Fault elimination</td><td>Use/Operation
(workplace)</td><td>Setting-up
(workplace)</td></tr>
<tr><td>X</td><td colspan="2">X</td><td></td><td></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/Hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
JR10 - Feed roller table to the cooling bed/induction heating system
JT10 to JT50 - Pinch roll unit induction heating system and cooling line
JZ20, JZ40 - Induction heating system
- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-J10-01 to D-J10-03) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Once work has been completed and when all persons have left the danger zone and the access doors (D-J10-01 to D-J10-03) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-J10-01 to D-J10-03).
- the doors to the relevant danger zone are then electrically locked again (with guard locking)by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.6.11 Rail head quenching and tempering (cooling modules and roller table in area of cooling modules)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J11-01 to D-J11-04</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Rail head quenching and tempering
(cooling modules and roller table in area of cooling modules)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Setting work on the cooling modules</td></tr>
<tr><td>during:</td><td>Servicing/ Maintenance</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td colspan="2"></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/Hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JR10</td><td colspan="4">- Feed roller table to the cooling bed/induction heating system</td></tr>
<tr><td></td><td>JR20</td><td colspan="4">- Cooling bed entry roller table</td></tr>
<tr><td></td><td>JT10, JT20, JT30, JT40, JT50</td><td colspan="4">- Pinch roll unit within area of entire cooling line</td></tr>
<tr><td></td><td>JM10, JM20, JM30, JM40, JM50, JM60, JM70, JM80</td><td colspan="4">- Cooling module 1 to cooling module 8</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td></td><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP5.1".
Before entering the danger zone, the relevant plant area must be switched off from the CP5.1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
Note: When working on the cooling modules
- the modules must be protected by appropriate mechanical means against accidentally lowering.
- the shut-off valve on the water feed lines must be secured against restarting.

On completion of the work and once all persons have left the danger zone
- the access doors (D-J11-01 to D-J11-04) must be locked again and
- the keys deposited in the CP5.1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.6.12 Equipment in the area of the cooling bed (level ± 0.0 mm)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J12-01 and D-J12-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in the area of the cooling bed (feed-in)
(level ± 0.0 mm)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>JR10</td><td colspan="4">- Feed roller table to the cooling bed/induction heating system</td></tr>
<tr><td>JT10, JT20, JT30, JT40, JT50</td><td colspan="4">- Pinch roll unit within area of entire cooling line</td></tr>
<tr><td>JR20</td><td colspan="4">- Cooling bed run-in roller table</td></tr>
<tr><td>JC10</td><td colspan="4">- Pre-bending equipment (entry transfer car)</td></tr>
<tr><td>JC20</td><td colspan="4">- Cooling bed 1</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP5.1".
Before entering the danger zone, the relevant plant area must be switched off from the CP5.1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-J12-01 and D-J12-02) must be locked again and
- the keys deposited in the CP5.1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J13-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in the area of the cooling bed (discharge)
(level ± 0.0 mm)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
JR30 - Cooling bed run-out roller table
JC20 - Cooling bed 1
- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP5".
Before entering the danger zone the plant area must be switched off from the CP5 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-J13-01) must be locked again and
- the keys deposited in the CP5 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.6.13 Equipment in the area of the cooling bed (level ± -3,500 (-6,000) mm)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-J14-01 to D-J14-07</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in the area of the cooling bed
(level -3,500 (-6,000) mm)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JR20</td><td colspan="4">- Cooling bed run-in roller table</td></tr>
<tr><td></td><td>JC10</td><td colspan="4">- Pre-bending equipment (entry transfer car)</td></tr>
<tr><td></td><td>JC20</td><td colspan="4">- Cooling bed 1</td></tr>
<tr><td></td><td>JB10</td><td colspan="4">- Cooling bed fans</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP5.1".
Before entering the danger zone, the relevant plant area must be switched off from the CP5.1 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-J14-01 to D-J14-07) must be locked again and
- the keys deposited in the CP5.1 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.7 Sub-sections 7 and 8 (equipment in the area of the straighteners)

4.6.3.7.1 Entry guide

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="8">D-K01-01 and D-K01-02</td></tr>
<tr><td>Access to:</td><td colspan="8">Danger zone
Entry guide</td></tr>
<tr><td>Access required for:</td><td colspan="8">-</td></tr>
<tr><td rowspan="2">during:</td><td colspan="2">Maintenance (service)/ Repair work</td><td colspan="2">Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td colspan="2">X</td><td colspan="2">X</td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="8">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td colspan="2">JR30</td><td colspan="7">- Cooling bed run-out roller table</td></tr>
<tr><td colspan="2">KE10</td><td colspan="7">- Entry guide</td></tr>
<tr><td colspan="2">KE20</td><td colspan="7">- Rail manipulator</td></tr>
<tr><td colspan="2">KF11/KF12
KF111/KF112</td><td colspan="7">- Straightener drives</td></tr>
<tr><td colspan="2">LR10</td><td colspan="7">- Roller table downstream of straightener</td></tr>
<tr><td colspan="2"></td><td colspan="7">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="4">Access door/sliding lattice grate</td><td colspan="2">Light curtain</td><td colspan="2" rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="3">Electrically</td><td colspan="2" rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td colspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td colspan="2">X</td><td colspan="2"></td><td colspan="2">-</td></tr>
<tr><td>Design:</td><td colspan="8">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-K01-01 and D-K01-02) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Once work has been completed and when all persons have left the danger zone and the access doors (D-K01-01 and D-K01-02) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-K01-01 and D-K01-02).
- the doors to the relevant danger zone are then electrically locked again (with guard locking)by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.7.2 Horizontal roller straightener and rail manipulator

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-K02-01 and D-K02-02 and light curtain LB-K02-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Equipment in area of horizontal roller straightener
and
rail manipulator</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>JR30</td><td colspan="4">- Cooling bed run-out roller table</td></tr>
<tr><td></td><td>KE10</td><td colspan="4">- Entry guide,</td></tr>
<tr><td></td><td>KE20</td><td colspan="4">- Rail manipulator</td></tr>
<tr><td></td><td>KF10</td><td colspan="4">- Horizontal roller straightener</td></tr>
<tr><td></td><td>KF110</td><td colspan="4">- Vertical roller straightener</td></tr>
<tr><td></td><td>LR10</td><td colspan="4">- Roller table downstream of straightener</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-K02-01 and D-K02-02) must be activated.
The doors to the danger zone can only be opened and/or the light curtain is deactivated and the danger zone may/can only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and all persons have left the danger zone again
- the access doors (D-K02-01 and D-K02-02) must be closed again.
- the in-charge person may activate the acknowledgement button located outside near the corresponding access door (D-K02-01 and D-K02-02) or near the light curtain (LB-K02-03)
- the doors to the relevant danger zone are then electrically locked (with guard locking) by the machine control system.
- the light curtain (LB-K02-03) is re-activated.
- the plant can be put back into operation.
Service: - Under the 'Service' function, all required movements are performed at reduced speed, in jogging mode.
- Movement of the various plant components is only possible using an 'enabling pushbutton' to be supplied.
- Access to the respective danger zone is only allowed to authorised and appointed persons.
- Operators and maintenance crews must have been trained and instructed with regard to special potential hazards.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to Sections 4.6.1 and 4.6.2.</td></tr>
</table>

4.6.3.7.3 Horizontal roller straightener (preparation for straightening roller change, changing manipulator)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-K03-01 and D-K03-02 (and light curtain LB-K02-03)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Changing manipulator</td></tr>
<tr><td>Access required for:</td><td colspan="5">Preparation for straightening roller change</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
KN10 - Changing manipulator, horiz. straightening rollers
- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the danger zone of the changing manipulator (Danger zone 2), the request button located in the vicinity of the corresponding door (D-K03-01 and D-K03-02) must be activated.
The doors to the danger zone of the changing manipulator may only be opened and the danger zone entered once all hazardous movements in the changing manipulator danger zone have stopped.
Condition:
-The light curtain LB-K02-03 is activated.
- The unit (changing manipulator) has been taken out of operation.
- operation (changing manipulator) has been transferred to the "Local Desk" (LDRS6)
Work may be carried out in the changing manipulator danger zone (Danger zone 2).

Once work has been completed and all persons have left the changing manipulator danger zone (Danger zone 2) and the access doors have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-K03-01 and D-K03-02).
- the doors to the corresponding changing manipulator danger zone (Danger zone 2) are electrically locked by the machine control system.
- the changing manipulator may be switched on again and put into operation.
When the doors are closed (D-K03-01 and D-K03-02) the light curtain (LB-K02-03) can/may be deactivated.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to Sections 4.6.1 and 4.6.2.</td></tr>
</table>

4.6.3.7.4 Horizontal roller straightener (Maintenance (servicing), set-up/straightening roller change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-K03-01 and D-K03-02 (and light curtain LB-K02-03)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
1. Changing manipulator
(access via door D-K03-01 and D-K03-02)
2. Horizontal roller straightener
(access via light curtain LB-K02-03)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Straightening roller change (work in danger zone)</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>KN10</td><td colspan="4">1. Changing manipulator (horizontal straightening rollers)
(with access via door D-K03-01 and D-K03-02)</td></tr>
<tr><td></td><td colspan="4">2. Horizontal roller straightener
(with access via light curtain LB-K02-03)</td></tr>
<tr><td>KE10</td><td colspan="4">- Entry guide,</td></tr>
<tr><td>KE20</td><td colspan="4">- Rail manipulator</td></tr>
<tr><td>KF10</td><td colspan="4">- Horizontal roller straightener</td></tr>
<tr><td>KF110</td><td colspan="4">- Vertical roller straightener</td></tr>
<tr><td>LR10</td><td colspan="4">- Straightener run-out roller table</td></tr>
<tr><td>JR30</td><td colspan="4">3. Cooling bed run-out roller table</td></tr>
<tr><td></td><td colspan="4">4. Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td>X</td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the changing manipulator danger zone (Danger zone 2), the request button located in the vicinity of the corresponding door (D-K03-01 and D-K03-02) must be activated.
The doors to the danger zone of the changing manipulator may only be opened and the danger zone entered once all hazardous movements in the changing manipulator danger zone have stopped.
Condition:
- the light curtain (LB-K02-03) is active or the straightener is switched off (if multiple access to the danger zone is required).
- the Automatic mode (changing manipulator) has been taken out of operation.
- operation of the changing manipulator has been transferred to the 'Local Desk' (LDRS6)
Work may be carried out in the changing manipulator danger zone (Danger zone 2).
To enter the danger zone (Danger zone 1) the request button located in the vicinity of the light curtain (LB-K02-03) must be activated.
Condition:
- The straightener has been taken out of operation.
Work can be carried out in the danger zone (Danger zone 1).
Once work has been completed and all persons have left the danger zone of the straightener, the in-charge person may activate the acknowledgement button located outside the danger zone in the area of the light curtain (LB-K02-03).
The straightener can be brought back into operation again.
Condition: The light curtain (LB-K02-03) is active.

<u>Description continued on following page:</u></td></tr>
</table>

Description continued:

Once work has been completed and **all persons** have left the danger zone of the **changing manipulator** again,

- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-K03-01 and D-K03-02).
- the doors to the corresponding danger zone are electrically locked by the machine control system.
- the **changing manipulator** may be switched on again and **put into operation**.

Special operating conditions:
For setting-up purposes it may be required that persons stay in the danger zone (**Danger zone 1/2**).

Prior to **entering** the danger zone (**Danger zone 2**) the request button located in the vicinity of the corresponding door (D-K03-01 and D-K03-02) must be activated.

As soon as the access has been granted from the control station '**CP5**', the control commands are transferred to the ' **Local Desk' (LDRSx)**. Upon actuation of the key-operated switch at the local control desk "**Local Desk**", all plant components in the danger zone (**Changing manipulator**) can be operated only from the '**Local Desk**' in the **Semi-automatic mode**. During this process, all movements are performed step by step and one after the other.

All operations performed step by step may be carried out only according to the instructions (radio communication) given by a further person who stays outside the range of movement.

The movements are indicated by an **optical** and **acoustic signal**.

Note: **Automatic operation** of the **changing manipulator** is only permitted once all persons have left the danger zone and the **doors** D-K03-01 and D-K03-02 are closed and **locked**.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to **Sections 4.6.1 and 4.6.2**.

4.6.3.7.5 Vertical roller straightener (changing manipulator, preparation for straightening roller change, set-up/straightening roller change)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-K04-01 and D-K04-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Vertical roller straightener
Changing manipulator</td></tr>
<tr><td>Access required for:</td><td colspan="5">Preparation for straightening roller change
Set-up/straightening roller change (work in danger zone)</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>JR30</td><td colspan="4">- Cooling bed run-out roller table</td></tr>
<tr><td>KE10</td><td colspan="4">- Straightener entry guide</td></tr>
<tr><td>KE20</td><td colspan="4">- Rail manipulator</td></tr>
<tr><td>KF10</td><td colspan="4">- Horizontal roller straightener</td></tr>
<tr><td>KF110</td><td colspan="4">- Vertical roller straightener</td></tr>
<tr><td>KF150</td><td colspan="4">- Changing manipulator, vert. straightening rollers</td></tr>
<tr><td>LR10</td><td colspan="4">- Straightener run-out roller table</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept closed during machine movement.
Prior to entering the danger zone (Danger zone 1) the request button located in the vicinity of the corresponding access door (D-K04-01 and D-K04-02) must be activated.
The doors to the danger zone can only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Special operating conditions:
For setting-up purposes (preparation for straightening roller change) it may be necessary for persons to remain in the danger zone (Danger zone 1).
- Under the 'Service' function, all required movements are performed at reduced speed, in jogging mode.
- Movement of the various plant components is only possible using an 'enabling pushbutton' to be supplied.
- Access to the respective danger zone is only allowed to authorised and appointed persons.
- Operators and maintenance crews must have been trained and instructed with regard to special potential hazards.
Once work has been completed and all persons have left the danger zone and the access doors (D-K04-01 and D-K04-02) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-K04-01 and D-K04-02).
- the doors to the danger zone concerned are again electrically closed by the machine control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.8 Sub-section 9 (equipment in area of transfer bed 1 upstream of test centre)

4.6.3.8.1 Straightener run-out roller table

Safety element (doors/light curtain):	**D-L01-01 to D-L01-10 and light curtain LB-L01-11**			
Access to:	Danger zones **Straightener run-out roller table**			
Access required for:	Scrap separation and removal			
during:	Maintenance (service)/ Repair work	Troubleshooting/ Fault elimination	Use/Operation (workplace)	Setting-up (workplace)
	X	**X**	**YES**	
Access conditions:	Drives/hydraulic and pneumatic movements safely switched off (The **applicable version of the motors and components list** must be observed):			
	JR30	- Cooling bed run-out roller table		
	KF10	- Horizontal roller straightener		
	KF110	- Vertical roller straightener		
	LR10	- Straightener run-out roller table		
	YZ27	- Travelling carriage		
	LR16	- Pusher		
		- Hydraulic/pneumatic equipment		

Design of the safety element (interlocking):	Access door/sliding lattice grate			Light curtain	
	Mechanically **locked**	Electrically **locked**	Electrically with **request/ acknowledgement button**	with **request/ acknowledgement button**	designed acc to: PLr (EN ISO 13849-1)
	X			**X**	**-**

Design:

The danger zone must be kept **locked** during machine movements.

1. **Access via doors D-L01-01 to D-L01-10**

   The keys for the access doors to the danger zone are deposited in the control room "**CP5**".

   Before entering the danger zone the **plant area must be switched off** from the **CP5** control room.

   The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once **all hazardous movements** in the danger zone have stopped.

   Work can be carried out in the danger zone (**Danger zone 1**).

   Access to this danger zone is permitted only for **authorised and designated personnel**.

   The user must assign the keys to the designated persons.

   On completion of the work and once **all persons** have left the danger zone
   - the access doors (D-L01-01 to D-L01-10) must be **locked** again and
   - the keys deposited in the **CP5** control room.

   The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

2. **Access via light curtain LB-L01-11**

   Prior to entering the danger zone (**Danger zone 1**) the request button located in the area of the corresponding access doors (LB-L01-11) must be actuated.

   The light curtain is deactivated and the danger zone can/may be entered only once **all hazardous movements** in the danger zone have stopped.

   Work can be carried out in the danger zone (**Danger zone 1**).

   Once work has been completed and **all persons** have left the danger zone again
   - the in-charge person may activate the acknowledgement button located outside near the light curtain (LB-L01-11).
   - the light curtain (LB-L01-11) is **re-activated**.
   - the plant can be put back **into operation**.

Description continued on following page:

Description continued:

Note: **Automatic operation** of the plant is only permitted once all persons have left the danger zone and the **doors** D-L01-01 to D-L01-10 have been closed and **locked** and the light curtain (LB-L01-11) has been **re-activated**.

Special operating conditions when **separating** and **removing** scrap:

For separating scrap it may be necessary for persons to remain in the danger zone (**Danger zone 1**).

- While work is being carried out (separation of rail sections) all drives (hydraulic and pneumatic movements) must be taken out of operation.
- Necessary movements of the **straightener run-out roller table** and/or the **travelling carriage** may only be carried out in manual mode.
- The operator must ensure that **no persons are in the danger zone (range of movement)** of the **straightener run-out roller table** and/or the **travelling carriage**
- The light curtain (LB-L01-11) is **deactivated** during movement of the **travelling carriage**.
- The movements of the **travelling carriage** must be announced by an **optical** and/or **acoustic signal**.
- Since the range of movement of the travelling carriage **cannot** be secured by a protective guard, the danger zone (area of movement) must be colour-marked (**yellow-black**)



.

All doors must be equipped with an escape unlocking mechanism on the inside.

For detailed descriptions on opening and closing of an electrically locked door as well as on activating and deactivating of a light curtain, please refer to **Sections 4.6.1 and 4.6.2**.

4.6.3.8.2 Area of transfer bed (with cross-transfer unit)

Safety element (doors/light curtain):	**D-L02-01 to D-L02-12**			
Access to:	Danger zones **Area of transfer bed** with **cross-transfer unit**			
Access required for:	-			
during:	Maintenance (service)/ Repair work	Troubleshooting/ Fault elimination	Use/Operation (workplace)	Setting-up (workplace)
	X	**X**		**-**

Access conditions:

Drives/hydraulic and pneumatic movements safely switched off
(The **applicable version of the motors and components list** must be observed):

LR10	- Straightener run-out roller table
LC10, LP10	- Cross-transfer unit upstream of test centre
LR20	- Roller table upstream of test centre 1
LR30	- Roller table downstream of test centre 1
LR40	- Roller table upstream of test centre 2
LR50	- Roller table downstream of test centre 2
LR70	- Visual inspection of return roller table
LZ10, LZ20	- Roller table within test centre 1 and 2
	- Hydraulic/pneumatic equipment

Design of the safety element (interlocking):

Access door/sliding lattice grate			Light curtain	
Mechanically **locked**	Electrically		with **request/ acknowledgement button**	designed acc to: PLr (EN ISO 13849-1)
	locked	with **request/ acknowledgement button**		
X				**-**

Design:

The danger zone must be kept **locked** during machine movements.

The user must assign the keys to the designated persons.

The keys for the access doors to the danger zone are deposited in the control room "**CP6**".

Before entering the danger zone, the **plant area must be switched off** from the **CP6** control room.

The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once **all hazardous movements** in the danger zone have stopped.

Work can be carried out in the danger zone (**Danger zone 1**).

Access to this danger zone is permitted only for **authorised and designated personnel**.

On completion of the work and once **all persons** have left the danger zone
- the access doors (D-L02-01 to D-L02-12) must be **locked** again and
- the keys deposited in the **CP6** control room.

The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.

4.6.3.9 Sub-section 10 (equipment in area of test centre and cold saws 1 to 4)

4.6.3.9.1 Cold saw 1 upstream of test centre 1

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-L03-01</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zone
Cold saw 1 upstream of test centre 1</td></tr>
<tr><td>Access required for:</td><td colspan="6">Preparation and changing of saw blades</td></tr>
<tr><td>during:</td><td colspan="2">Maintenance (service)/ Repair work</td><td colspan="2">Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td colspan="2">X</td><td colspan="2">X</td><td></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LR20</td><td colspan="5">- Roller table upstream of test centre 1</td></tr>
<tr><td>LS10</td><td colspan="5">- Cold saw 1 upstream of test centre 1</td></tr>
<tr><td>LQ10</td><td colspan="5">- Tilting table at cold saw 1</td></tr>
<tr><td>LR30</td><td colspan="5">- Roller table downstream of test centre 1</td></tr>
<tr><td>LZ10</td><td colspan="5">- Roller table inside test centre 1</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td colspan="2" rowspan="3">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L03-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone, especially those of the cold saws, have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the cold saw:
- the cold saw must have been taken out of operation
- the control commands for the cold saw must have been transferred from the CP6 control room to the 'Local Desk' (LDCS1).

Once work has been completed and all persons have left the danger zone and the access door (D-L03-01) has been closed,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door (D-L03-01) .
- the control commands for the saw are transferred again from the Local Desk (LDCS1) to the CP6 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.
The cold saw can be brought back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.9.2 Deburring and brushing unit (in area of cold saw 1)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L07-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Deburring and brushing unit
(in area of cold saw 1)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR20</td><td colspan="4">- Roller table upstream of test centre 1</td></tr>
<tr><td></td><td>LQ10</td><td colspan="4">- Tilting table at cold saw 1</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Pinch roll unit upstream of deburring and brushing unit (in area of cold saw 1)</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Deburring and brushing unit (in area of cold saw 1)</td></tr>
<tr><td></td><td>LR30</td><td colspan="4">- Roller table downstream of test centre 1</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Roller table inside test centre 1</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L07-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Condition:
When working on the deburring and brushing unit the deburring and brushing unit must be taken out of operation secured against re-starting.
When working on the extraction system the extraction system must be switched off and secured against re-starting.
Once work has been completed and when all persons have left the danger zone and the access door (D-L07-01) has been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-L07-01).
- the doors to the relevant danger zone are once again electrically locked by the machine control system.
The deburring and brushing unit as well as the extraction system can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.9.3 Access to container cellar at deburring and brushing unit (in area of cold saw 1)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L14-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Access to container cellar
at deburring and brushing unit
(in area of cold saw 1)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Handling and emptying of containers</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Deburring and brushing unit (in area of cold saw 1)</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Extraction system at deburring and brushing unit
(in area of cold saw 1)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP6".
Before entering the danger zone, the plant area must be switched off from the CP6 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-L14-01) must be locked again and
- the keys deposited in the CP6 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.9.4 Cold saw 2 downstream of test centre 1

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L04-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Cold saw 2 downstream of test centre 1</td></tr>
<tr><td>Access required for:</td><td colspan="5">Preparation and changing of saw blades</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR20</td><td colspan="4">- Roller table upstream of test centre 1</td></tr>
<tr><td></td><td>LR30</td><td colspan="4">- Roller table downstream of test centre 1</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Roller table inside test centre 1</td></tr>
<tr><td></td><td>LZ10</td><td colspan="4">- Pinch roll unit upstream of deburring and brushing unit (in area of cold saw 1)</td></tr>
<tr><td></td><td>LS20</td><td colspan="4">- Cold saw 2 downstream of test centre 1</td></tr>
<tr><td></td><td>LQ20</td><td colspan="4">- Tilting table at cold saw 2</td></tr>
<tr><td></td><td>LP20</td><td colspan="4">- Disappearing stopper upstream of test centre 1 (must be raised)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L04-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone, especially those of the cold saws, have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the cold saw:
- the cold saw must have been taken out of operation
- the control commands for the cold saw must have been transferred from the CP7 control room to the 'Local Desk' (LDCS2).

Once work has been completed and when all persons have left the danger zone and the access door (D-L04-01) has been closed again,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door (D-L04-01) .
- the control commands for the saw are transferred again from the Local Desk (LDCS2) to the CP7 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.
The cold saws can be brought back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions on opening and closing an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.9.5 Cold saw 3 upstream of test centre 2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L05-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Cold saw 3 upstream of test centre 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">Preparation and changing of saw blades</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LR40</td><td colspan="5">- Roller table upstream of test centre 2</td></tr>
<tr><td>LS30</td><td colspan="5">- Cold saw 3 upstream of test centre 2</td></tr>
<tr><td>LQ30</td><td colspan="5">- Tilting table at cold saw 3</td></tr>
<tr><td>LR50</td><td colspan="5">- Roller table downstream of test centre 2</td></tr>
<tr><td>LZ20</td><td colspan="5">- Roller table inside test centre 2</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L05-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone, especially those of the cold saws, have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the cold saw:
- the cold saw must have been taken out of operation
- the control commands for the cold saw must have been transferred from the CP6 control room to the
'Local Desk' (LDCS3).

Once work has been completed and all persons have left the danger zone and the access door (D-L05-01) has been closed,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door (D-L05-01).
- the control commands for the saw are transferred again from the Local Desk (LDCS3) to the CP6 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.
The cold saws can be brought back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.9.6 Deburring and brushing unit (in area of cold saw 3)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L08-01 to D-L08-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Deburring and brushing unit
(in area of cold saw 3)</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR40</td><td colspan="4">- Roller table upstream of test centre 2</td></tr>
<tr><td></td><td>LQ30</td><td colspan="4">- Tilting table at cold saw 3</td></tr>
<tr><td></td><td>LZ20</td><td colspan="4">- Pinch roll unit upstream of deburring and brushing unit (in area of cold saw 3)</td></tr>
<tr><td></td><td>LZ20</td><td colspan="4">- Deburring and brushing unit (in area of cold saw 3)</td></tr>
<tr><td></td><td>LR50</td><td colspan="4">- Roller table downstream of test centre 2</td></tr>
<tr><td></td><td>LZ20</td><td colspan="4">- Roller table inside test centre 2</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L08-01 to D-L08-03) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Condition:
When working on the deburring and brushing unit the deburring and brushing unit must be taken out of operation secured against re-starting.
When working on the extraction system the extraction system must be switched off and secured against re-starting.
Once work has been completed and when all persons have left the danger zone and the access doors (D-L08-01 to D-L08-03) have been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding door (D-L08-01 to D-L08-03).
- the doors to the relevant danger zone are once again electrically locked by the machine control system.
The deburring and brushing unit as well as the extraction system can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.9.7 Access to container cellar at deburring and brushing unit (in area of cold saw 3)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L15-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Access to container cellar
at deburring and brushing unit
(in area of cold saw 3)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Handling and emptying of containers</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td></td><td></td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LZ20</td><td colspan="5">- Deburring and brushing unit (in area of cold saw 3)</td></tr>
<tr><td>LZ20</td><td colspan="5">- Extraction system at deburring and brushing unit
(in area of cold saw 3)</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td>Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP6".
Before entering the danger zone, the plant area must be switched off from the CP6 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-L15-01) must be locked again and
- the keys deposited in the CP6 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.9.8 Cold saw 4 downstream of test centre 2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L06-01</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Cold saw 4 downstream of test centre 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">Preparation and changing of saw blades</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR40</td><td colspan="4">- Roller table upstream of test centre 2</td></tr>
<tr><td></td><td>LR50</td><td colspan="4">- Roller table downstream of test centre 2</td></tr>
<tr><td></td><td>LZ20</td><td colspan="4">- Roller table inside test centre 2</td></tr>
<tr><td></td><td>LZ20</td><td colspan="4">- Pinch roll unit upstream of deburring and brushing unit (in area of cold saw 3)</td></tr>
<tr><td></td><td>LS40</td><td colspan="4">- Cold saw 4 downstream of test centre 2</td></tr>
<tr><td></td><td>LQQ0</td><td colspan="4">- Tilting table at cold saw 4</td></tr>
<tr><td></td><td>LP40</td><td colspan="4">- Disappearing stopper upstream of test centre 2 (must be raised)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L06-01) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone, especially those of the cold saws, have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the cold saw:
- the cold saw must have been taken out of operation
- the control commands for the cold saw must have been transferred from the CP7 control room to the 'Local Desk' (LDCS4).

Once work has been completed and all persons have left the danger zone and the access door (D-L06-01) has been closed,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door (D-L06-01).
- the control commands for the saw are transferred again from the Local Desk (LDCS4) to the CP7 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.
The cold saws can be brought back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.9.9 Points of access to test centre 1 and 2

The doors must be kept **closed** during production.

Access to these areas is allowed only to **trained personnel** who have been **authorised** by the plant owner**.**

The keys to the danger zones are kept in the respective control room of the allocated plant section.

The owner must assign the keys to the designated persons, and the authorisation to hold keys must be stipulated in the operating and working instructions.

The access and switch-off conditions concerning the relevant access doors to the danger zone of the test centres are stipulated by the supplier in the risk analysis.

4.6.3.9.10 Return roller table (12.3) visual inspection (in area of test centre 2)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L09-01 to D-L09-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Points of access to return roller table in area of test centre 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LR70</td><td colspan="4">- Visual inspection of return roller table</td></tr>
<tr><td></td><td colspan="4"></td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to:
PLr
(EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP6/7.
Before entering the danger zone, the plant area must be switched off from the CP6/7 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-L09-01 to D-L09-03) must be locked again and
- the keys deposited in the CP6/7 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.9.11 Roller table downstream of test centre 2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L10-01 to D-L10-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Points of access to roller table downstream of test centre 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR40</td><td colspan="4">- Roller table upstream of test centre 2</td></tr>
<tr><td></td><td>LZ20</td><td colspan="4">- Roller table inside test centre 2</td></tr>
<tr><td></td><td>LR50</td><td colspan="4">- Roller table downstream of test centre 2</td></tr>
<tr><td></td><td>LS40</td><td colspan="4">- Cold saw 4 downstream of test centre 2</td></tr>
<tr><td></td><td>LQQ0</td><td colspan="4">- Tilting table at cold saw 4</td></tr>
<tr><td></td><td>LP40</td><td colspan="4">- Disappearing stopper upstream of test centre 2 (must be raised)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP6/7.
Before entering the danger zone, the plant area must be switched off from the CP6/7 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-L10-01 to D-L10-02) must be locked again and
- the keys deposited in the CP6/7 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.10 Sub-section 11 (equipment in area of transfer bed 2, visual inspection and cold saws 5 to 10)

4.6.3.10.1 Rail cross-transfer unit downstream of test centre 1 and 2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L11-01 to D-L11-20</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Rail cross-transfer unit downstream of test centre 1 and 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td colspan="2"></td><td>-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LR20, LR40, LZ10, LZ20</td><td colspan="5">- Roller tables upstream of and in test centres 1 and 2</td></tr>
<tr><td>LR30</td><td colspan="5">- Roller table (9.5) downstream of test centre 1</td></tr>
<tr><td>LR50</td><td colspan="5">- Roller table (9.7) downstream of test centre 2</td></tr>
<tr><td>LC30, LC40</td><td colspan="5">- Rail cross-transfer unit downstream of test centre 1 and 2</td></tr>
<tr><td>LR60</td><td colspan="5">- Feed roller table (12.1) visual inspection</td></tr>
<tr><td>LR70</td><td colspan="5">- Return roller table (12.3) visual inspection</td></tr>
<tr><td>PR10, PR20</td><td colspan="5">- Roller tables downstream of saws 5 to 10</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP7.
Before entering the danger zone the relevant plant area must be switched off from the CP7 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-L11-01 to D-L11-20) must be locked again and
- the keys deposited in the CP7 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.10.2 Points of access to workstations, visual inspection

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">---</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zone
Equipment in area of visual inspection</td></tr>
<tr><td>Access required for:</td><td colspan="6">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td></td><td colspan="2">YES</td><td colspan="2"></td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="6">The points of access to the visual inspection workstations are mechanically closed.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP8".
Access to the danger zone must be limited to those persons who are directly engaged in the operation of the structural steel rolling mill and its auxiliary equipment.
Access to this danger zone is permitted only for authorised and designated personnel.
All persons entering or leaving the danger zone must sign in and out in the CP8 control room, in order that the plant operator knows there are persons inside the danger zone (Workstation visual inspection).
On completion of the work and once all persons have left the danger zone, the access doors must again be closed and the keys deposited in control room CP8.
Note: - Do not cross the check lines during operation
- For your own safety at least one emergency stop pushbutton must be fitted at each check point.
- The danger zone around the check lines must be colour-marked (yellow-marked). 
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.10.3 Roller tables upstream of cold saws 5 to 10 and return roller table

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L11-21 to D-L11-26 or D-L11-27</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Roller tables upstream of cold saws 5 to 10 and return roller table</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LR61, PR11, PR21</td><td colspan="3">- Roller tables in area of line 1</td><td colspan="2">(D-L11-21)</td></tr>
<tr><td>LR62, PR12, PR22</td><td colspan="3">- Roller tables in area of line 2</td><td colspan="2">(D-L11-22)</td></tr>
<tr><td>LR63, PR13, PR23</td><td colspan="3">- Roller tables in area of line 3</td><td colspan="2">(D-L11-23)</td></tr>
<tr><td>LR64, PR14, PR24</td><td colspan="3">- Roller tables in area of line 4</td><td colspan="2">(D-L11-24)</td></tr>
<tr><td>LR65, PR15, PR25</td><td colspan="3">- Roller tables in area of line 5</td><td colspan="2">(D-L11-25)</td></tr>
<tr><td>LR66, PR16, PR26</td><td colspan="3">- Roller tables in area of line 6</td><td colspan="2">(D-L11-26)</td></tr>
<tr><td>LR70, PR41</td><td colspan="3">- Return roller table (12.3) visual inspection</td><td colspan="2">(D-L11-27)</td></tr>
<tr><td></td><td colspan="3">- Hydraulic/pneumatic equipment</td><td colspan="2"></td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP8".
Before entering the danger zone, the relevant plant area must be switched off from the CP8 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-L11-21 to D-L11-26 or D-L11-27) must be locked again and
- the keys deposited in the CP8 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.10.4 Cold saws 5 to 7 (line 1 to 3)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="6">D-L12-01 to D-L12-05</td></tr>
<tr><td>Access to:</td><td colspan="6">Danger zone
Cold saws 5 to 7 (line 1 to 3)</td></tr>
<tr><td>Access required for:</td><td colspan="6">Preparation and changing of saw blades</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td colspan="2"></td><td colspan="2">YES</td></tr>
<tr><td>Access conditions:</td><td colspan="6">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR60</td><td colspan="5">- Feed roller tables (12.1), visual inspection (line 1 to 3)</td></tr>
<tr><td></td><td>PR10, PR20</td><td colspan="5">- Feed roller table (12.1), visual inspection (line 1 to 3)</td></tr>
<tr><td></td><td>LS50, LS60, LS70</td><td colspan="5">- Cold saws 5 to 7 (line 1 to 3)</td></tr>
<tr><td></td><td>LQ50, LQ60, LQ70</td><td colspan="5">- Tilting tables at cold saws 5 to 7 (line 1 to 3)</td></tr>
<tr><td></td><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="6">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L12-01 to D-L12-05) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone, especially those of the cold saws, have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the cold saw:
- the cold saw must have been taken out of operation
- the control commands for the cold saw must have been transferred from the CP8 control room to the 'Local Desk' (LDCS5, LDCS6 or LDCS7).

Once work has been completed and all persons have left the danger zone and the access door (D-L12-01 to D-L12-05) has been closed,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door (D-L12-01 to D-L12-05) .
- the control commands for the saw are transferred again from the 'Local Desk' (LDCS5, LDCS6 or LDCS7) to the CP8 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.
The cold saws can be brought back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.10.5 Cold saws 8 to 10 (line 4 to 6)

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-L13-01 to D-L13-05</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zone
Cold saws 8 to 10 (line 4 to 6)</td></tr>
<tr><td>Access required for:</td><td colspan="5">Preparation and changing of saw blades</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td colspan="2"></td><td>YES</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td></td><td>LR60</td><td colspan="4">- Feed roller tables (12.1), visual inspection (line 4 to 6)</td></tr>
<tr><td></td><td>PR10, PR20</td><td colspan="4">- Feed roller table (12.1), visual inspection (line 4 to 6)</td></tr>
<tr><td></td><td>LS80, LS90, LS100</td><td colspan="4">- Cold saws 8 to 10 (line 4 to 6)</td></tr>
<tr><td></td><td>LQ80, LQ90, LQ100</td><td colspan="4">- Tilting tables at cold saws 8 to 10 (line 4 to 6)</td></tr>
<tr><td></td><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zones must be kept locked during production.
Prior to entering the danger zone (Danger zone 1) the request button arranged in the vicinity of the respective door (D-L13-01 to D-L13-05) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone, especially those of the cold saws, have stopped.
Work can be carried out in the danger zone (Danger zone 1).

Condition:
For work on the cold saw:
- the cold saw must have been taken out of operation
- the control commands for the cold saw must have been transferred from the CP8 control room to the 'Local Desk' (LDCS8, LDCS8 or LDCS10).

Once work has been completed and all persons have left the danger zone and the access door (D-L13-01 to D-L13-05) has been closed,
- the person in charge may press one of the acknowledgement buttons located outside next to the relevant door (D-L13-01 to D-L13-05) .
- the control commands for the saw are transmitted again from the 'Local Desk' (LDCS8, LDCS9 or LDCS10) to the CP8 control room.
- the doors to the relevant danger zone are again electrically locked by the machine control system.
- the plant can be put into operation again.
The cold saws can be brought back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.</td></tr>
</table>

4.6.3.11 Sub-section 12 (equipment in area of transfer bed 3 and straightening presses)

4.6.3.11.1 Rail cross-transfer unit downstream of cold saws 5 to 10

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-P01-01 to D-P01-07</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Rail cross-transfer unit downstream of cold saws 5 to 10</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>LR60</td><td colspan="5">- Roller tables (12.1) in area of transfer bed 3</td></tr>
<tr><td>LR70</td><td colspan="5">- Return roller table (12.3) visual inspection</td></tr>
<tr><td>PR40</td><td colspan="5">- Roller tables (13.1) upstream of straightening presses 1 and 2</td></tr>
<tr><td>PC10</td><td colspan="5">- Rail cross-transfer unit (transfer bed 3) downstream of visual inspection</td></tr>
<tr><td>PP10</td><td colspan="5">- Disappearing stopper downstream of cold saws (13m)</td></tr>
<tr><td>PP20</td><td colspan="5">- Disappearing stopper downstream of cold saws (26m)</td></tr>
<tr><td>PP30</td><td colspan="5">- Disappearing stopper downstream of cold saws (65m)</td></tr>
<tr><td>PP40</td><td colspan="5">- Disappearing stopper downstream of cold saws (130m)</td></tr>
<tr><td>PR10, PR20</td><td colspan="5">- Roller tables in transfer bed</td></tr>
<tr><td>QR11 and QR12</td><td colspan="5">- Roller tables downstream of straightening presses 1 and 2</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td>Mechanically locked</td><td colspan="2">Electrically</td><td>with request/ acknowledgement button</td></tr>
<tr><td></td><td>locked</td><td>with request/ acknowledgement button</td><td></td><td></td><td></td></tr>
<tr><td>X</td><td></td><td></td><td></td><td colspan="2">-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in the control room "CP8".
Before entering the danger zone, the relevant plant area must be switched off from the CP8 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
On completion of the work and once all persons have left the danger zone
- the access doors (D-P01-01 to D-P01-07) must be locked again and
- the keys deposited in the CP8 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.11.2 Access points in area of straightening presses 1 & 2

Safety element (doors/light curtain):	**D-P02-01 to D-P02-03**			
Access to:	Danger zones **Access points in area of straightening presses 1 & 2**			
Access required for:	-			
during:	Maintenance (service)/ Repair work	Troubleshooting/ Fault elimination	Use/Operation (workplace)	Setting-up (workplace)
	X	**X**	**YES 1)**	
Access conditions:	Drives/hydraulic and pneumatic movements safely switched off (The **applicable version of the motors and components list** must be observed):			
	PR40	- Roller table (13.1) upstream of straightening presses 1 and 2		
	PR40	- Roller table (13.1) in area of straightening presses 1 and 2		
	PZ30	- Straightening press 1 and 2		
	QR10	- Roller tables downstream of straightening presses		
		- Hydraulic/pneumatic equipment		

Design of the safety element (interlocking):	Access door/sliding lattice grate			Light curtain	
	Mechanically **locked**	Electrically		with **request/ acknowledgement button**	designed acc to: PLr (EN ISO 13849-1)
		locked	with request/ acknowledgement button		
	X				**-**

Design:

The danger zone must be kept **locked** during machine movements.

The user must assign the keys to the designated persons.

The keys for the access doors to the danger zone are deposited in the control room "**CP8**".

Before entering the danger zone, the relevant **plant area must be switched off** from the **CP8** control room.

The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once **all hazardous movements** in the danger zone have stopped.

Work can be carried out in the danger zone (**Danger zone 1**).

Access to this danger zone is permitted only for **authorised and designated personnel**.

Note:
- When performing service and maintenance work, all drives must be switched off and secured against re-starting.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.

On completion of the work and once **all persons** have left the danger zone
- the access doors (D-P02-01 to D-P02-03) must be **locked** again and
- the keys deposited in the **CP8** control room.

The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.

All doors must be equipped with an escape unlocking mechanism on the inside.

1) Operation of straightening presses 1 and 2 is performed in **Manual** mode from the local control desks **LDGP1** and **LDGP2**.

4.6.3.12 Sub-section 13 (equipment in the processing area for 130m rails)

4.6.3.12.1 Rail cross-transfer unit downstream of straightening presses 1 & 2

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-Q01-01 to D-Q01-03</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Rail cross-transfer unit downstream of straightening presses 1 & 2</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td colspan="2">Use/Operation (workplace)</td><td>Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td colspan="2"></td><td>-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>PR40</td><td colspan="5">- Roller table (13.1) upstream of straightening presses 1 and 2</td></tr>
<tr><td>PR40</td><td colspan="5">- Roller table (13.1) in area of straightening presses 1 and 2</td></tr>
<tr><td>QR10</td><td colspan="5">- Roller table (14.1) downstream of straightening presses 1 and 2</td></tr>
<tr><td>QC20</td><td colspan="5">- Rail cross-transfer unit (14.4) for short rails</td></tr>
<tr><td>RR10</td><td colspan="5">- Roller table (15.1) for short rails</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td>Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP9.
Before entering the danger zone, the relevant plant area must be switched off from the CP9 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
Note: - When performing service and maintenance work, all drives must be switched off and secured against re-starting.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.
On completion of the work and once all persons have left the danger zone
- the access doors (D-Q01-01 to D-Q01-03) must be locked again and
- the keys deposited in the CP9 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.12.2 Loading area for 130m rails

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-Q02-01 to D-Q02-09</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Loading area for 130m rails</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>PR40</td><td colspan="4">- Roller table (13.1) upstream of straightening presses 1 and 2</td></tr>
<tr><td>PR40</td><td colspan="4">- Roller table (13.1) in area of straightening presses 1 and 2</td></tr>
<tr><td>QR10</td><td colspan="4">- Roller table (14.1) downstream of straightening presses 1 and 2</td></tr>
<tr><td>QC10</td><td colspan="4">- Rail cross-transfer unit (14.2) for loading the 130m rails</td></tr>
<tr><td></td><td colspan="4">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP9.
Before entering the danger zone, the relevant plant area must be switched off from the CP9 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
Note: - When performing service and maintenance work, all drives must be switched off and secured against re-starting.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.
On completion of the work and once all persons have left the danger zone
- the access doors (D-Q02-01 to D-Q02-09) must be locked again and
- the keys deposited in the CP9 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.13 Sub-section 14 (equipment in short rail processing area)

4.6.3.13.1 Roller table (15.1) for short rails

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-R01-01 to D-R01-02</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Roller table (15.1) for short rails</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
RR10 - Roller table (15.1) for short rails
- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td></td><td></td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP9.
Before entering the danger zone, the relevant plant area must be switched off from the CP9 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
Note: - When performing service and maintenance work, all drives must be switched off and secured against re-starting.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.
On completion of the work and once all persons have left the danger zone
- the access doors (D-R01-01 to D-R01-02) must be locked again and
- the keys deposited in the CP9 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.13.2 Rail cross-transfer unit for short rails

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-R02-01 to D-R02-08</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Rail cross-transfer unit (15.2) for short rails</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td>during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td></td><td>X</td><td>X</td><td></td><td colspan="2">-</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):</td></tr>
<tr><td>RR10</td><td colspan="5">- Roller table (15.1) for short rails</td></tr>
<tr><td>RC10</td><td colspan="5">- Rail cross-transfer unit (15.2) for short rails and pusher</td></tr>
<tr><td>RR20</td><td colspan="5">- Loading roller table (15.3) for short rails</td></tr>
<tr><td>RP10</td><td colspan="5">- Aligning stopper</td></tr>
<tr><td></td><td colspan="5">- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td></tr>
<tr><td>X</td><td></td><td></td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
The user must assign the keys to the designated persons.
The keys for the access doors to the danger zone are deposited in control room CP9.
Before entering the danger zone, the relevant plant area must be switched off from the CP9 control room.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Access to this danger zone is permitted only for authorised and designated personnel.
Note: - When performing service and maintenance work, all drives must be switched off and secured against re-starting.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.
On completion of the work and once all persons have left the danger zone
- the access doors (D-R02-01 to D-R02-08) must be locked again and
- the keys deposited in the CP9 control room.
The plant may be started only after the plant operator has made sure that all keys have been deposited in the control room again.
All doors must be equipped with an escape unlocking mechanism on the inside.</td></tr>
</table>

4.6.3.13.3 Drilling machines 1 and 2 for short rails

<table>
<tr><td>Safety element (doors/light curtain):</td><td colspan="5">D-R03-01 (or D-R04-01)</td></tr>
<tr><td>Access to:</td><td colspan="5">Danger zones
Drilling machines 1 and 2 for short rails</td></tr>
<tr><td>Access required for:</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">during:</td><td>Maintenance (service)/ Repair work</td><td>Troubleshooting/ Fault elimination</td><td>Use/Operation (workplace)</td><td colspan="2">Setting-up (workplace)</td></tr>
<tr><td>X</td><td>X</td><td></td><td colspan="2">YES 1)</td></tr>
<tr><td>Access conditions:</td><td colspan="5">Drives/hydraulic and pneumatic movements safely switched off
(The applicable version of the motors and components list must be observed):
RR20 - Loading roller table (15.3) for short rails
RS20 - Drilling machines 1 and 2
RC20 - Loading bed 1 for short rails
RC30 - Loading bed 2 for short rails
- Hydraulic/pneumatic equipment</td></tr>
<tr><td rowspan="3">Design of the safety element (interlocking):</td><td colspan="3">Access door/sliding lattice grate</td><td>Light curtain</td><td rowspan="2">designed acc to: PLr (EN ISO 13849-1)</td></tr>
<tr><td rowspan="2">Mechanically locked</td><td colspan="2">Electrically</td><td rowspan="2">with request/ acknowledgement button</td></tr>
<tr><td>locked</td><td>with request/ acknowledgement button</td><td>X</td><td>X</td><td></td><td>-</td></tr>
<tr><td>Design:</td><td colspan="5">The danger zone must be kept locked during machine movements.
Prior to entering the danger zone (Danger zone 1) the request button located in the area of the corresponding access door (D-R03-01 (or D-R04-01)) must be activated.
The doors to the danger zone can/may only be opened and the danger zone can/may only be entered once all hazardous movements in the danger zone have stopped.
Work can be carried out in the danger zone (Danger zone 1).
Note: - When performing service and maintenance work, all drives must be switched off and secured against re-starting.
- For functions with limited safety, consideration must be given to the conditions indicated in Section 4.5, Definition and relationships of operating modes, functions and operating situations.

Once work has been completed and when all persons have left the danger zone and the access door (D-R03-01 (or D-R04-01)) has been closed again,
- the in-charge person may activate the acknowledgement button arranged outside near the corresponding access door (D-R03-01 (or D-R04-01)).
- the doors to the relevant danger zone are then electrically locked again (with guard locking) by the machine's control system.
- the plant can be put back into operation.
All doors must be equipped with an escape unlocking mechanism on the inside.
For detailed descriptions for opening and closing of an electrically locked door, please refer to Section 4.6.1.

1) To operate drilling machines 1 and 2 in Manual mode, automatic mode is taken out of operation and the control functions are transferred to the corresponding local control desk
LDD1 (for drilling machine 1)
or
LDD2 (for drilling machine 2)
.</td></tr>
</table>

4.6.3.14 Sub-section 15 (equipment in area of transfer car)

Since, for operational reasons, the area of movement (danger zone) of the transfer cars cannot be secured with a protective guard or railing, and therefore persons may be within the range of movement (danger zone) of the transfer cars as the cars are being moved, the transfer cars may only be operated under the following conditions:

1. On the bay floor, the area of movement (danger zone) of the transfer car must be provided with yellow-black colour marks.



2. Remaining within the danger zone (area inside the yellow and black colour marks) while the machine is operational is **prohibited**.
3. Movement of the **transfer cars** must be announced by an **optical** (flashing light) and/or **acoustic signal** (horn).
4. Necessary movements of the **transfer cars** may only be executed
   - from the local desk **LDFA1** or **LDFA2**
     (in the case of transfer cars in the area of the tandem mill stand or transfer bed 1)
     as well as from the local desk**LDFA3**
     (in the straightener run-out roller table downstream of the straightener (see also Section 4.6.3.8.1))
     in manual mode (**jog mode**).
   - from the **co-travelling** control station **LDSA1**
     (in the case of bloom transfer cars in the area "upstream of the furnace")
     in manual mode (**jog mode**).
5. The area of movement (danger zone) must be visible to the operator.
6. The operator must ensure that no persons remain within the area of movement (danger zone).

4.7 Explanatory notes on danger-zone safeguarding

4.7.1 General

The danger zones of the rail rolling mill with the machinery/components/plants included in it must be safeguarded by an access control facility (guard (protecting fence)) preventing access (also unintended access) to the danger zones.

The design must comply with the requirements of EN 953 and with the safety measures specified in EN ISO 13857, EN 349.

The necessary fence heights were selected according to EN ISO 13857, Table 2, with the height of the protecting fence depending on the distance from and height in relation to the danger zone.

The height of the protective fence must be at least 1,400 mm. Depending on the hazardous situation, fence heights of between 1,800 and 2,000 mm may be required. The fence heights are given in the layout drawings.

In various plant sections, the danger zones, with all of the machinery/components/plants included in them, must be secured with railings so as to prevent access (also unintended access) to the danger zones.
The design and implementation must comply with the specifications laid out in DIN EN 14133-3.
The height of the railings must be 1,100 mm.

The access doors required here, unless otherwise protected by an access control system (safety switch with guard locking), must be secured by keys. The danger zones must be kept **locked** during production.

Access to these danger zone is permitted only for **authorised and designated personnel**.

The user must assign the keys to the designated persons.



Access to the danger zones must be limited to those persons who are directly engaged in the operation of the rail rolling mill and its auxiliary equipment and who are able to assess the potential hazards. The personnel must have been trained and instructed in safety issues.

The authorised persons shall be appointed by the owner.

Staying in the danger zones is **not** permitted.

The range of movement/danger zone of the entire plant can be viewed by the operator, or (if necessary) is covered by cameras and displayed on monitors in the control room.

An intercom system or radio communication is used for communication between the persons involved.



In some areas, access is blocked by existing machinery/components, control cabinets, furnaces, etc. These items must not be removed unless a suitable substitute is provided.

No claim for the supply of such items may be derived from the descriptions and representations.

4.7.2 Relevant standards

DIN EN ISO 12100	Safety of machinery, Basic concepts, General principles for design.
DIN EN ISO 14121-1	Safety of machinery, Risk assessment Part 1: Principles
DIN EN 349	Safety of machinery Minimum distances to avoid crushing of parts of the human body.
DIN EN 953	Safety of machinery General requirements for separating guards
DIN EN 1088	Safety of machinery, Interlocking devices associated with guards; Principles for design and selection
DIN EN ISO 13857	Safety of machinery, Safety distances to prevent hazard zones from being reached by lower and upper limbs
DIN EN 1037	Safety of machinery, Isolation from power supply and dissipation of stored energy, Prevention of unexpected start-up

4.7.3 Emergency-stop functions

4.7.3.1 Stop category 0 according to DIN EN ISO 13850

The emergency-stop buttons on the electrical control cabinets must be designed as emergency-stop buttons of Stop category 0, i.e., the power supply to the machine is interrupted immediately.

4.7.3.2 Stop category 1 according to DIN EN ISO 13850

Various emergency stop buttons are arranged on the machine/plant. After pressing one of the emergency stop buttons, the drives of the respective plant component come to a standstill (decelerated). The drives are then switched off (power supply off/depressurised).

This also applies when a light curtain as well as the escape unlocking mechanism of an electrically locked door is triggered.

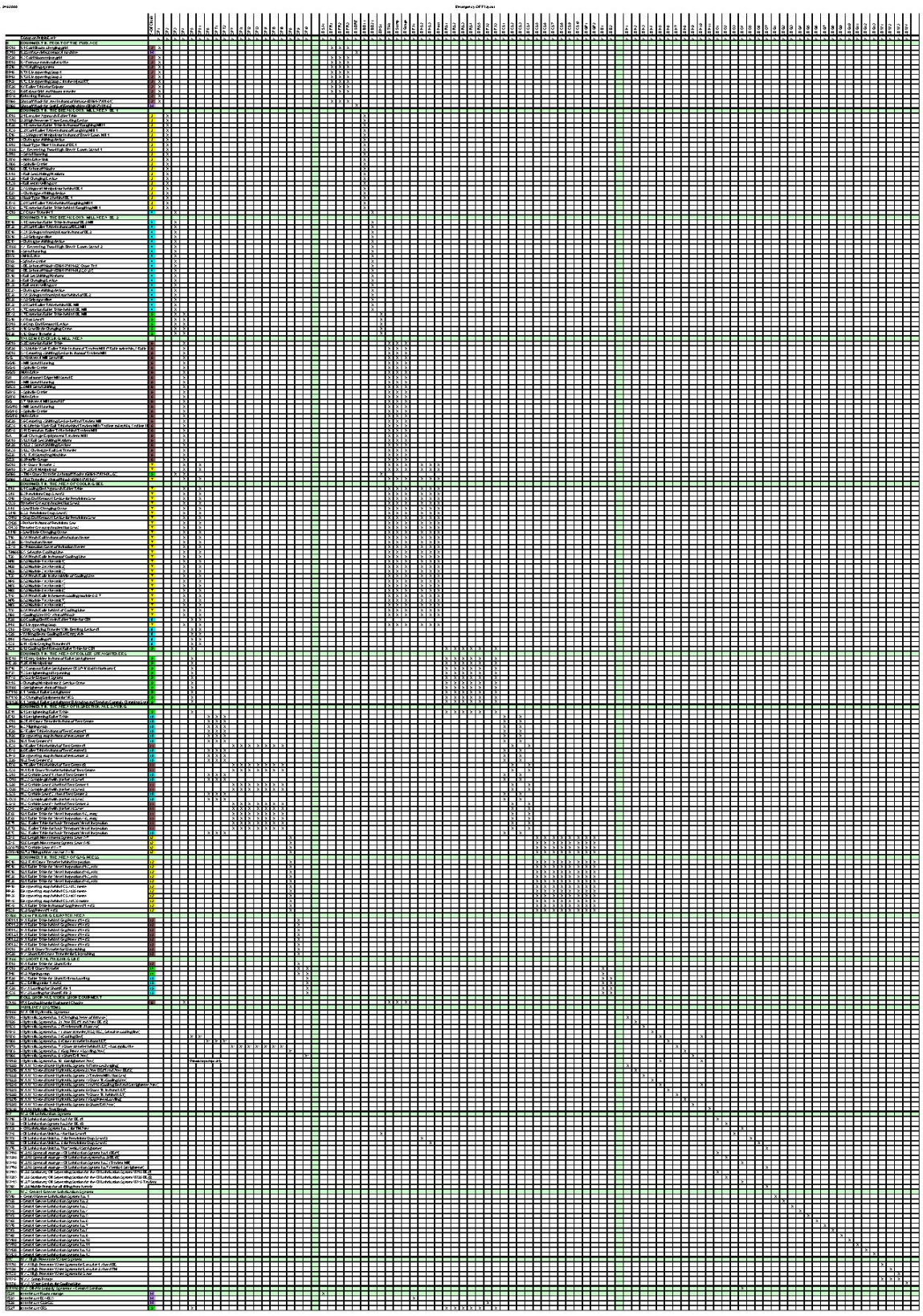


The hydraulic accumulator is not depressurised in the event of the emergency-stop function being triggered.

The pressure oil supply to the machine is merely interrupted.

Individual lines/hydraulic cylinders may still be pressurised even when the power supply is switched off (resulting from a weight force).

4.7.3.3 Emergency stop circuits



(For details see "Other relevant documents")

Figure 4-3 Emergency stop circuits (E-STOP overview)

4.7.4 Battery limit

In the safety layout, the limits of scope of supply of the SMS Meer supply portion are represented as a broken green line. This represents a demarcation line to a danger zone which due to the existing danger potential must be safeguarded before the start of production (if necessary or possible) by a railing so as to prevent access (including unintended access) to the danger zones.

Due to the local conditions, the installation of railings at the battery limit is not absolutely necessary if access to the relevant danger zones concerned is effectively prevented by other machine or plant sections or structural measures.

If the material flows across this limit of scope of supply, it must be ensured that material transportation is always absolutely safe.

In detail this means:

- Material transport only in the event of a release signal.
- Transport drive stop, when the emergency stop function has been tripped.



In all cases, access across the limits of scope of supply to the danger zones in question must be reliably prevented prior to starting production.

The person with overall responsibility for the erection of the machine/plant is responsible for the safeguarding measures.

4.7.5 Sealing of foundation

The foundation must be sealed oil-proof. The sealing coat must be inspected and/or renewed at the usual intervals.



If the foundation design is such that contamination of the ground water is ruled out, there is no need for oil-proof sealing.

The national regulations and laws constitute the basis for the execution and, if necessary, for the implementation of a test.

4.7.6 Assignment of key

Various danger zones / access doors are locked by keys. The danger zones must be kept **locked** during production.

The key for the access doors to the danger zones are in the control room "**CP…**" of the respective "section".

The user must assign the keys to the designated persons.

4.7.7 Authorised personnel

The rail rolling mill has been designed and manufactured to conform to the currently applicable standards and guidelines. Nevertheless, hazards during machine operation, setup, tool change, service and maintenance cannot be fully avoided. For this reason, the rail rolling mill must be operated and serviced only by trained personnel who are familiar with the plant and the residual hazards.

If several operators and/or auxiliary personnel are working at the machine, a person with overall responsibility must be appointed.

4.7.8 Escape and rescue routes

The arrangement and identification of the escape and rescue routes is the responsibility of the machine and plant user. The requirements for this are usually defined in country-specific ordinances on workplaces and in various standards and directives.

Sufficiently large access areas, which can also be used as escape and rescue routes, if necessary, must be provided at all doors in the danger zone safeguards and at all control and maintenance stations.

4.7.9 Emergency exit

Emergency exits from the machine cellar/hydraulic rooms must be marked accordingly. Exit hatches must not be obstructed or blocked by objects which would restrict or prevent their use.

The floor opening must be safeguarded such that no objects or persons may fall into it.

4.7.10 Existing hazards, residual risks

Residual risks are those risks existing on the plant that cannot be eliminated by any appropriate measures without substantially reducing the machine functions.

The residual risks are described in the list of possible residual risks and/or in the operating instructions. This document must be made available to all persons working with and on the machine.

The operating personnel must be instructed with regard to the special potential hazards (on the basis of the list of possible residual risks).

Personnel must be instructed at regular intervals.

4.8 List of figures

Figure 4-1 Definition of plant components/sub-sections (Part 1) 9
Figure 4-2 Definition of plant components/sub-sections (Part 2) 10
Figure 4-3 Emergency stop circuits (E-STOP overview) 110

4.9 Other relevant documents

Drawing	Designation
	Sub-section 1 (equipment in area upstream of furnace)
0100133-00.650_01	Layout
0100133-00.650_02	Layout of slab transfer car area
	Sub-section 2 (equipment in area of descaler)
0100133-00.651_01	Layout
	Sub-section 3 (equipment in area of stand BD1)
0100133-00.651_01	Layout
0100133-00.651_02	Horizontal sections
	Sub-section 4 (equipment in area of stand BD2)
0100133-00.652_01	Layout
0100133-00.652_02	Horizontal section
	Sub-section 5 (equipment in the area of tandem mill stand)
0100133-00.653_01	Layout
	Sub-section 6 (equipment in the area of the cooling bed)
0100133-00.654_01	Layout
0100133-00.654_02	
0100133-00.654_03	
0100133-00.654_04	Horizontal section
0100133-00.654_05	
0100133-00.654_06	Sections
	Sub-sections 7 and 8 (equipment in area of straighteners)
0100133-00.655_01	Layout
0100133-00.655_02	Horizontal section
0100133-00.655_03	Detail
	Sub-section 9 (equipment in area of transfer bed 1 upstream of test centre)
0100133-00.656_01	Layout
	Sub-section 10 (equipment in area of test centre and cold saws 1 to 4)
0100133-00.657_01	Layout

Drawing	Designation
	Sub-section 11 (equipment in area of transfer bed 2, visual inspection and cold saws 5 to 10)
0100133-00.658_01	Layout
	Sub-section 12 (equipment in area of transfer bed 3 and straightening presses)
0100133-00.659_01	Layout
	Sub-section 13 (equipment in the processing area for 130m rails)
0100133-00.660_01	Layout
0100133-00.654_06	Sections
	Sub-section 14 (equipment in short rail processing area)
0100133-00.661_01	Layout
0100133-00.661_02	Sections

Applicable version of the motors and components list

Document		Issued
MuK_RAILBHIL_rev_E.xls	Motors and components list	20.12.2011
0100133-01.002_01_E_Off_a.pdf	(E-Stop overview)	16.01.2012
B_AA04 E Stop_Table_B.xls	(E-Stop matrix)	16.01.2012